

Arithmetic Quantum Mechanics

Lab Book

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1 Overview

This campaign's north star is a general definition and workflow that assigns to an arbitrary arithmetic scheme one or more *canonical quantum systems*: kinematics realised as a Hilbert space, observables realised as an algebra, and, where possible, dynamics realised as automorphisms or as projective unitary representations of symmetry groups, with fusion categories as the expected general endpoint. The word *canonical* is a technical requirement here, not a mood: the assignment must be exhibited as a functor, every choice on which it depends must be named explicitly, and a theorem must state exactly what is independent of those choices.

The concrete guiding path fixes a target before the generalities are attempted: the prime field $\mathbb{F}_p$ of characteristic p , its canonical symplectic space $\mathbb{F}_p \times \mathbb{F}_p$, and the finite Weyl–Heisenberg system quantizing it. Every general definition this campaign proposes must reproduce this case exactly, and is tested against it before anything else. The result is recorded in this lab book, one topic per section, each statement carrying its full hypotheses, an honest status, and its provenance (§1); nothing in these pages is asserted at a strength its evidence does not support.

Status

The argument DAG `claims/CLAIMS.md` currently carries **28** rows, all issued by the campaign's first increment (`briefs/wh-kappa-target.md`): the canonical quantum system attached to a finite field κ , with every choice it depends on named. Of these, **22** are **PROVED** (four of them **PROVED**[†], meaning the proved statement is restricted to $p = 2$ or to odd p and displays that restriction in its own text — never abbreviated into the label), **4** are **SKETCH** (an argument exists and is written down, but has not yet been through this campaign's hostile-critic round), **1** is **CONJECTURE**, and **1** is **REFUTED** (with its surviving weaker statement recorded in the adjacent row). Every definition D1–D11 of `definitions.md` and every row below is restated in full, under a descriptive name, in the sections that follow; the table records only where.

The increment's central finding, proved in §4, is that the construction depends on a second datum beyond the field and the additive character: a *polarizing cocycle*. At odd characteristic this choice is immaterial and has a canonical representative. At characteristic two it is not immaterial: the admissible cocycles split into exactly two $SL_2(\kappa)$ -orbits, distinguished by the Arf invariant of an associated quadratic form, giving two genuinely non-isomorphic Heisenberg groups — a fact detectable by an operator identity inside the quantum system itself. Per `PRD.md`, a sharp result of this shape is a positive structural statement about the definition, not a negative one.

Claim id	Statement (short)	Status	Where
WH-FORM	The symplectic form is κ -bilinear, alternating, nondegenerate; its κ -linear and $\mathbb{F}_p$ -linear isometries coincide, both $SL_2(\kappa)$.	PROVED	Thm. 2.6
WH-COMM	The Weyl commutation relations hold uniformly in p , for every admissible cocycle.	PROVED	Thm. 2.7
WH-ALG	The twisted algebra is simple, dimension q^2 , centre $\mathbb{C}$, for every cocycle.	PROVED	Thm. 2.8
WH-POL	Every line is Lagrangian; a Schrödinger model is a pair (L, χ) , and all such pairs give isomorphic modules.	PROVED	Thm. 3.3

continued on next page

Claim id	Statement (short)	Status	Where
WH-POL-2	At $p = 2$, characters induced on a line with $Q_\beta \neq 0$ are forced to order 4; no such forcing at odd p .	PROVED [†]	Thm. 3.4
WH-SVN	Uniqueness of the finite Weyl representation: one irreducible unitary representation of dimension q , intertwiner unique up to $U(1)$, every automorphism inner.	PROVED	Thm. 3.5
WH-ALG-MAT	The algebra is isomorphic to $M_q(\mathbb{C})$, by a non-canonical isomorphism.	PROVED	Thm. 3.6
WH-BETA-a	Admissible polarizing cocycles form a torsor of size q^3 under the symmetric bilinear forms.	PROVED	Thm. 4.2
WH-BETA-b	At odd p , the cocycle is immaterial: $\omega/2$ is a canonical representative fixing an explicit isomorphism between any two choices.	PROVED [†]	Thm. 4.3
WH-BETA-c	At $p = 2$, no symmetric cocycle exists: the odd- p route to a canonical choice is unavailable.	PROVED [†]	Thm. 4.4
WH-BETA-d	At $p = 2$, Q_β is a quadratic form with polar form ω ; the resulting element-order profile of $H_\beta(\kappa)$ is computed, and at least two isomorphism classes occur.	PROVED [†]	Thm. 4.5
WH-BETA-TYPE	Heart of the increment. The Arf invariant classifies $\text{Adm}(\omega)$ into exactly two $SL_2(\kappa)$ -orbits at $p = 2$, both nonempty, giving exactly two isomorphism classes of Heisenberg group.	SKETCH	Thm. 4.7
WH-BETA-EPS	The Arf type is detected inside the quantum system itself, by an explicit operator identity.	SKETCH	Thm. 4.8
WH-BETA-LEVEL	The algebra and its Weyl frame do not depend on β at any characteristic; the finite level- μ_p frame does, and only at $p = 2$.	SKETCH	Thm. 4.10
WH-WEIL-a	Frame automorphisms surject onto $SL_2(\kappa)$ with kernel $\hat{V}$, at every characteristic.	PROVED	Thm. 5.2
WH-WEIL-b	At odd p , this extension splits explicitly, giving a projective unitary representation of $SL_2(\kappa)$.	PROVED [†]	Thm. 5.3
WH-WEIL-c	At $p = 2$, μ_2 -valued phases occur exactly over the orthogonal subgroup $O(Q_\beta)$, proper for the reference cocycle.	PROVED [†]	Thm. 5.4

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Claim id	Statement (short)	Status	Where
WH-WEIL-d	At $p = 2$, μ_4 is forced exactly when $O(Q_\beta)$ is proper — always except the single case $q = 2$, anisotropic.	PROVED [†]	Thm. 5.5
WH-WEIL-e	Whether the extension splits at $p = 2$ for every $q = 2^m$; verified only at $q = 2, 4$.	CONJECTURE	Conj. 5.6
WH-SYMM	The algebra does not remember the κ -module structure of $V(\kappa)$; its intrinsic $\mathbb{F}_p$ -linear symmetry is the strictly larger $Sp_{2m}(\mathbb{F}_p)$ once $m > 1$.	PROVED	Thm. 7.1
WH-CHOICE	Beyond κ , exactly two data: ψ and β ; the resulting algebras are always isomorphic, but no isomorphism among them is distinguished.	PROVED	Thm. 6.2
WH-CANON	The projective Hilbert space is canonical in (κ, ψ) , at fixed β .	PROVED	Thm. 6.3
WH-CANON-BETA	The projective Hilbert space is canonical independently of β as well.	SKETCH	Thm. 6.4
WH-FUNCT-a	The trace-normalized character family is not restriction-compatible along field extensions; trivial exactly when $p \mid n$.	PROVED	Thm. 8.1
WH-FUNCT-b	$(\kappa, \psi) \mapsto A_{\psi, \beta_0}(V(\kappa))$ is a functor on character-compatible embeddings.	PROVED	Thm. 8.2
WH-FUNCT-b-SEC	<i>Refuted:</i> a Frobenius-compatible character exists for every finite field.	REFUTED	Thm. 8.3
WH-FUNCT-c	The Frobenius obstruction, exactly: invariance forces $c \in \mathbb{F}_p$, and $\mathbb{F}_{p^p}$ then restricts trivially to the prime field.	PROVED	Thm. 8.4
WH-FUNCT-d	Compatible character systems over an algebraic closure are exactly the restrictions of its characters nontrivial on $\mathbb{F}_p$.	PROVED	Thm. 8.5

2 The symplectic object of a finite field, and its twisted operator algebra

Throughout the labbook p denotes a prime, $m \geq 1$ an integer, $q := p^m$, and $\kappa := \mathbb{F}_q$ the finite field of order q , with prime field $\mathbb{F}_p \subseteq \kappa$. **The case $p = 2$ is in scope everywhere in this campaign:** no definition or theorem below is stated only for odd p and silently assumed to extend; where a statement holds only for $p = 2$ or only for odd p this is marked in the statement itself, never left to a reader's guess. This section builds the ground floor of the construction — the symplectic vector space attached to κ , the two choices (a polarizing cocycle and an additive character) the quantum system is built from, the twisted operator algebra those choices produce, and its associated Heisenberg group — and proves the three structural facts that hold *uniformly* in the characteristic and for *every* admissible choice of cocycle. What is and is not canonical about the choices themselves is the subject of later sections, in particular §4.

2.1 The symplectic space of a finite field

Definition 2.1 (The symplectic object attached to a finite field). For a finite field κ , put $V(\kappa) := \kappa \oplus \kappa$, and define

$$\omega : V(\kappa) \times V(\kappa) \rightarrow \kappa, \quad \omega((a, b), (a', b')) := ab' - a'b.$$

Scope. This is a stipulation: a vector space and a bilinear pairing written down by formula. On its own it asserts nothing about ω — not that it is κ -bilinear, not that it is alternating, not that it is nondegenerate. Those are properties, established as Theorem 2.6 below. No restriction on the characteristic is built into the definition; $p = 2$ is included on the same footing as every other prime.

Provenance. ids: D1; proved in: —; tested by: —; reviewed in: —.

2.2 Polarizing cocycles: the datum β , introduced

The Weyl operators of §2.4 below are built not from ω directly but from a bilinear map whose antisymmetrization is ω . Choosing one such map is a second datum of the construction, on exactly the same footing as the choice of additive character ψ of §2.3. This subsection only introduces the datum and its notation; the fact that it forms a torsor, that it is immaterial at odd characteristic, and that it splits the construction into two genuinely different isomorphism types at $p = 2$ is the content of Theorems 4.2–4.7 in §4, and is not anticipated here.

Definition 2.2 (Admissible polarizing cocycles, and the reference cocycle). Fix a finite field κ and its symplectic object $(V(\kappa), \omega)$ of Definition 2.1. The set of *admissible polarizing cocycles* is

$$\text{Adm}(\omega) := \{ \beta : V(\kappa) \times V(\kappa) \rightarrow \kappa \mid \beta \text{ is } \kappa\text{-bilinear and } \beta - \beta^T = \omega \},$$

where $\beta^T(v, v') := \beta(v', v)$. A *polarizing cocycle* is a choice of element $\beta \in \text{Adm}(\omega)$, and it is a **datum of the construction**, on the same footing as the character ψ of Definition 2.3 — not a convention fixed once and for all. The map

$$\beta_0((a, b), (a', b')) := ab'$$

lies in $\text{Adm}(\omega)$ and is called the *reference cocycle*: a label for a point of the torsor of Definition 2.2, not a canonical point of it (Theorem 4.2).

Scope. Nothing here asserts that $\text{Adm}(\omega)$ is nonempty, that it has any particular size, that β_0 is a typical or a distinguished element of it, or that the choice of β is or is not immaterial to anything built from it. All of that is claimed, and proved, later: Theorems 4.2–4.7 of §4 are where this datum’s structure is actually established. In particular the tempting odd-characteristic habit of writing $\omega/2$ for “the” polarizing cocycle is not part of this definition and is not available at $p = 2$, where division by 2 is division by 0; §4 makes this precise.

Provenance. ids: D2; proved in: —; tested by: —; reviewed in: —.

2.3 The phase datum

Definition 2.3 (Phase datum, and the trace-normalized family). A *phase datum* for κ is a nontrivial additive character $\psi : (\kappa, +) \rightarrow \mathbb{C}^\times$, i.e. a group homomorphism with $\psi \neq 1$. Write $X(\kappa)$ for the set of such ψ . Inside $X(\kappa)$, the *trace-normalized family* is defined as follows: for $\kappa = \mathbb{F}_{p^m}$ put

$$\text{Tr}_{\kappa/\mathbb{F}_p}(z) := z + z^p + z^{p^2} + \cdots + z^{p^{m-1}},$$

and for ζ a primitive p -th root of unity in $\mathbb{C}$, put

$$\psi_\zeta(z) := \zeta^{\text{Tr}_{\kappa/\mathbb{F}_p}(z)}.$$

Scope. Two choices are named here and neither is suppressed: the character ψ itself (an arbitrary element of $X(\kappa)$), and, when working inside the distinguished family, the root of unity ζ . The absolute trace $\text{Tr}_{\kappa/\mathbb{F}_p}$ is canonical — it is the same $\mathbb{F}_p$ -linear map regardless of any further choice — but ψ_ζ is canonical only once ζ is fixed, and §8 shows in detail how far a compatible choice of ψ across a tower of fields can and cannot be made.

Provenance. ids: D3; proved in: —; tested by: —; reviewed in: —.

2.4 Weyl operators, the twisted algebra, and the Heisenberg group

Definition 2.4 (Weyl operators and the twisted observable algebra). Fix a phase datum ψ as in Definition 2.3 and a polarizing cocycle $\beta \in \text{Adm}(\omega)$ as in Definition 2.2. The *twisted operator algebra* $A_{\psi,\beta}(V)$ is the $\mathbb{C}$ -vector space

$$A_{\psi,\beta}(V) := \bigoplus_{v \in V(\kappa)} \mathbb{C} \cdot W_\beta(v)$$

with the product fixed, once and for all, on basis elements by

$$W_\beta(v) W_\beta(v') := \psi(\beta(v, v')) W_\beta(v + v').$$

Scope. The ordering of the product on the right ($\psi(\beta(v, v'))$, not $\psi(\beta(v', v))$) is fixed as part of the definition and is not symmetrized; which of the two orderings a later stipulated model (such as M_0 of §3) actually realizes is a computation, not a convention, and is checked explicitly there. $A_{\psi,\beta}(V)$ depends on κ , ψ and β and on nothing else; that this dependence is associative, unital and of dimension q^2 is Theorem 2.7(a) below, not part of the stipulation.

Provenance. ids: D4; proved in: —; tested by: —; reviewed in: —.

Definition 2.5 (The Heisenberg group of a polarizing cocycle). With $\beta \in \text{Adm}(\omega)$ as above, the *Heisenberg group of β* is the set $H_\beta(\kappa) := \kappa \times V(\kappa)$ with product

$$(t, v)(t', v') := (t + t' + \beta(v, v'), v + v').$$

Scope. This construction uses no character ψ ; it is a group attached to κ and β alone. It does depend on β , and — this is the content of §4 — at $p = 2$ its isomorphism class as an abstract group depends on more than the isomorphism class of the pair $(V(\kappa), \omega)$: it depends on which point of the torsor $\text{Adm}(\omega)$ was chosen. Nothing about that dependence is claimed or used here.

Provenance. ids: D5; proved in: —; tested by: —; reviewed in: —.

2.5 The form is bilinear, alternating, nondegenerate, and its isometries

PROVED

Theorem 2.6 (The symplectic form of a finite field, at every characteristic). *Let κ be any finite field, $p = 2$ included, with $V(\kappa)$ and ω as in Definition 2.1. Then:*

- (a) ω is κ -bilinear;
- (b) $\omega(v, v) = 0$ for every $v \in V(\kappa)$ — alternating in the strong sense, not merely antisymmetric;
- (c) ω is nondegenerate: if $\omega(v, \cdot) \equiv 0$ then $v = 0$;

- (d) every $\beta \in \text{Adm}(\omega)$ (Definition 2.2) is a 2-cocycle for the abelian group $(V(\kappa), +)$, i.e. $\beta(v, v') + \beta(v + v', v'') = \beta(v, v'' + v) + \beta(v', v'')$ for all v, v', v'' — equivalently $\beta(v, v') + \beta(v + v', v'') = \beta(v', v'') + \beta(v, v' + v'')$ — and $\beta - \beta^T = \omega$;
- (e) the group of κ -linear automorphisms g of $V(\kappa)$ with $\omega(gv, gv') = \omega(v, v')$ for all v, v' is exactly $SL_2(\kappa)$;
- (f) the group of merely $\mathbb{F}_p$ -linear automorphisms g of $V(\kappa)$ (i.e. additive maps that need not commute with multiplication by κ) with $\omega(gv, gv') = \omega(v, v')$ for all v, v' is also exactly $SL_2(\kappa)$ — the larger class of symmetries one might a priori expect contains nothing beyond the κ -linear ones.

Scope. Part (f) is a statement about the κ -valued form ω itself; it is emphatically *not* a statement about the $\mathbb{F}_p$ -valued form $\psi \circ \omega$ obtained by composing with a character, whose isometries are the strictly larger group $Sp_{2m}(\mathbb{F}_p)$ once $m > 1$ — that is Theorem 7.1 of §7. Nothing here addresses which $\beta \in \text{Adm}(\omega)$ is chosen, or classifies $\text{Adm}(\omega)$ itself; that is §4.

Provenance. ids: D1, D2, WH-FORM; proved in: theory/wh-kappa.md §1; tested by: theory/checks/wh_kappa_check.py (C1,C2); theory/checks/fp-symmetry.py (S1); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. (a) Fix $v = (a, b)$, $v' = (a', b')$. Each argument of ω occurs exactly once in the formula $ab' - a'b$, and κ is commutative, so scaling either argument by $c \in \kappa$ scales the value by c , and additivity in each slot is distributivity of multiplication over addition in κ . This holds verbatim at $p = 2$: no step divides by 2 or by anything else.

(b) $\omega((a, b), (a, b)) = ab - ab = 0$, an identity in the commutative ring κ that cancels two literally equal terms, not two terms related by a factor of 2. It therefore holds at $p = 2$ exactly as it holds elsewhere. This is worth pausing on: antisymmetry alone, $\omega(v, v') = -\omega(v', v)$, gives only $2\omega(v, v) = 0$ on setting $v' = v$, which is vacuous once $2 = 0$. So at $p = 2$, (b) is a strictly stronger statement than antisymmetry and must be — and here is — verified by the direct computation above, not inferred from it.

(c) If $\omega(v, \cdot) \equiv 0$ for $v = (a, b)$ then in particular $\omega((a, b), (0, 1)) = a = 0$ and $\omega((a, b), (1, 0)) = -b = 0$, so $v = 0$.

(d) $\beta - \beta^T = \omega$ is the definition of $\text{Adm}(\omega)$ (Definition 2.2). For the cocycle identity: β is bi-additive (it is κ -bilinear, hence in particular additive in each argument), and for any bi-additive β both sides of the cocycle identity expand, term by term, to $\beta(v, v') + \beta(v, v'') + \beta(v', v'')$; the identity holds without using anything about ω beyond bi-additivity of β .

(e) Write $v = (a, b)$, $v' = (a', b')$ as the columns of a 2×2 matrix over κ ; then $\omega(v, v') = ab' - a'b = \det[v \mid v']$ by direct expansion. For $g \in GL_2(\kappa)$, $[gv \mid gv'] = g[v \mid v']$ as matrices, and multiplicativity of the determinant for 2×2 matrices over a commutative ring (itself a polynomial identity verified by direct expansion, holding in every characteristic) gives $\omega(gv, gv') = \det(g) \cdot \omega(v, v')$. Hence $\omega \circ g = \omega$ iff $(\det g - 1)\omega \equiv 0$; since $\omega((1, 0), (0, 1)) = 1 \neq 0$, this holds iff $\det g = 1$, i.e. iff $g \in SL_2(\kappa)$.

(f) Let $g : V(\kappa) \rightarrow V(\kappa)$ be additive (not assumed κ -linear) with $\omega(gv, gu) = \omega(v, u)$ for all v, u .

- *g is bijective.* If $gv = 0$ then $\omega(v, u) = \omega(gv, gu) = 0$ for every u , so $v = 0$ by (c); thus g is injective, hence bijective since $V(\kappa)$ is finite.
- *g sends the standard basis to a κ -basis.* Let $v_1 = (1, 0)$, $v_2 = (0, 1)$, so $\omega(v_1, v_2) = 1$. Then $\omega(gv_1, gv_2) = \omega(v_1, v_2) = 1 \neq 0$; were gv_1, gv_2 κ -dependent, bilinearity and (b) would force $\omega(gv_1, gv_2) = 0$ (if, say, $gv_2 = c \cdot gv_1$ then $\omega(gv_1, gv_2) = c\omega(gv_1, gv_1) = 0$ by (a) and (b)). So $\{gv_1, gv_2\}$ is κ -independent, hence a basis of the 2-dimensional space $V(\kappa)$.

- g is κ -linear. Fix $c \in \kappa$, $u \in V(\kappa)$, and put $w := g(cu) - c \cdot g(u)$ (subtraction makes sense because g is additive, hence $\mathbb{Z}$ -linear, even before κ -linearity is known). For $i = 1, 2$,

$$\omega(gv_i, w) = \omega(gv_i, g(cu)) - c\omega(gv_i, gu) = \omega(v_i, cu) - c\omega(v_i, u) = c\omega(v_i, u) - c\omega(v_i, u) = 0,$$

using the hypothesis $\omega(g\cdot, g\cdot) = \omega(\cdot, \cdot)$ for the middle equality and κ -bilinearity (a) for the outer ones. Since $\{gv_1, gv_2\}$ is a basis, $\omega(gv_1, \cdot)$ and $\omega(gv_2, \cdot)$ have trivial common kernel only at 0 by nondegeneracy (c) applied after a change of basis, so $w = 0$. Hence $g(cu) = c \cdot g(u)$ for all c, u : g is κ -linear.

By κ -linearity and (e), $\det g = 1$, so $g \in SL_2(\kappa)$; the converse containment is immediate from (e). This closes (f). $\square$

Remark. Part (f) forecloses a tempting shortcut: one might hope to characterize the $\mathbb{F}_p$ -linear isometries of ω as the “ ω -self-adjoint $\mathbb{F}_p$ -endomorphisms,” by analogy with the real symplectic case. This is false at $p = 2$: exhaustive enumeration finds 8 and 64 ω -self-adjoint $\mathbb{F}_p$ -endomorphisms of $V(\kappa)$ at $q = 2, 4$ respectively, against only 2 and 4 κ -scalars — the self-adjointness route simply does not isolate $SL_2(\kappa)$ once $p = 2$, and the direct argument above is what closes (f) instead. Independently, complete backtracking search over the $\mathbb{F}_p$ -linear isometries of the κ -valued form finds exactly $|SL_2(\kappa)| = 6, 24, 60, 120, 504, 720$ of them at $q = 2, 3, 4, 5, 8, 9$, matching an independent enumeration of $SL_2(\kappa)$ set-for-set at every one of these values (theory/checks/fp-symmetry.py, gates S1 and S4).

2.6 Weyl relations, uniformly in the characteristic and for every choice of cocycle

PROVED

Theorem 2.7 (The Weyl commutation relations, for every admissible cocycle). *Let κ be any finite field, ψ a phase datum (Definition 2.3), and $\beta \in \text{Adm}(\omega)$ an arbitrary admissible polarizing cocycle (Definition 2.2). Then, inside $A_{\psi, \beta}(V)$ of Definition 2.4:*

- (a) $A_{\psi, \beta}(V)$ is a unital associative $\mathbb{C}$ -algebra of dimension q^2 , with unit $W_\beta(0)$;
- (b) every $W_\beta(v)$ is invertible, with $W_\beta(v)^{-1} = \psi(\beta(v, v)) W_\beta(-v)$;
- (c) $W_\beta(v) W_\beta(v') = \psi(\omega(v, v')) W_\beta(v') W_\beta(v)$ for all $v, v' \in V(\kappa)$;
- (d) $W_\beta(u) W_\beta(v) W_\beta(u)^{-1} = \psi(\omega(u, v)) W_\beta(v)$ for all $u, v \in V(\kappa)$.

Scope. No part of this theorem uses which admissible cocycle $\beta \in \text{Adm}(\omega)$ was chosen, nor any property of β beyond bi-additivity and the defining relation $\beta - \beta^T = \omega$; in particular nothing here distinguishes the reference cocycle β_0 from any other point of the torsor. No hypothesis on the characteristic is used beyond what already went into Theorem 2.6. What *is* β -dependent — the isomorphism class of the group $H_\beta(\kappa)$ itself, at $p = 2$ — is not addressed here.

Provenance. ids: D1, D2, D3, D4, WH-FORM, WH-COMM; proved in: theory/wh-kappa.md §2; tested by: theory/checks/wh_kappa_check.py (C3, C4); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. (a) Associativity on basis elements amounts to

$$(W_\beta(v)W_\beta(v'))W_\beta(v'') = W_\beta(v)(W_\beta(v')W_\beta(v'')),$$

which unwinds to the 2-cocycle identity for $\psi \circ \beta$; this holds because β is a 2-cocycle (Theorem 2.6(d)) and ψ is multiplicative. Since $\beta(0, v) = \beta(v, 0) = 0$ (κ -bilinearity), and $\psi(0) = 1$, $W_\beta(0)$ is a two-sided unit. The dimension is $|V(\kappa)| = q^2$ by construction.

(b) By bi-additivity, $\beta(v, -v) = -\beta(v, v)$, so $W_\beta(v)W_\beta(-v) = \psi(\beta(v, -v))W_\beta(0) = \psi(-\beta(v, v)) \cdot 1$, and symmetrically on the other side; since ψ takes values in $\mathbb{C}^\times$, this exhibits a two-sided inverse.

(c) By the defining product, $W_\beta(v)W_\beta(v') = \psi(\beta(v, v'))W_\beta(v + v')$ and $W_\beta(v')W_\beta(v) = \psi(\beta(v', v))W_\beta(v + v')$. By (b), $W_\beta(v + v')$ is invertible, so

$$W_\beta(v)W_\beta(v') = \psi(\beta(v, v') - \beta(v', v)) W_\beta(v')W_\beta(v) = \psi(\omega(v, v')) W_\beta(v')W_\beta(v),$$

the last step by the defining relation $\beta - \beta^T = \omega$ of Definition 2.2.

(d) Multiply (c) on the right by $W_\beta(v)^{-1}$, whose existence is (b). $\square$

Remark (The centre of the Heisenberg group, and the algebra as a group-algebra quotient). The group $H_\beta(\kappa)$ of Definition 2.5 is indeed a group: associativity is the cocycle identity of Theorem 2.6(d), the identity is $(0, 0)$, and (t, v) has inverse $(-t + \beta(v, v), -v)$. Its centre is exactly $\kappa \times 0$: (t, v) is central iff $\beta(v, v') - \beta(v', v) = 0$ for every v' , i.e. iff $\omega(v, \cdot) \equiv 0$, i.e. iff $v = 0$ by Theorem 2.6(c). Consequently $A_{\psi, \beta}(V)$ is the quotient of the group algebra $\mathbb{C}[H_\beta(\kappa)]$ by the two-sided ideal identifying the central element $(t, 0)$ with the scalar $\psi(t)$: the algebra is a family over $\psi \in X(\kappa)$ at *fixed* β , and the group $H_\beta(\kappa)$ itself depends on β — a dependence that, Theorem 4.7 shows, is not always trivial once $p = 2$.

2.7 Simplicity of the twisted algebra

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Theorem 2.8 (Central simplicity of the twisted algebra, with no polarization). *With κ , ψ and $\beta \in \text{Adm}(\omega)$ as in Theorem 2.7, the algebra $A_{\psi, \beta}(V)$ of Definition 2.4 is simple, has centre $\mathbb{C} \cdot 1$, and $\dim_{\mathbb{C}} A_{\psi, \beta}(V) = q^2$.*

Scope. This theorem uses no choice of Lagrangian line, no module, and no property of β beyond Theorem 2.7; in particular it holds for every $\beta \in \text{Adm}(\omega)$, uniformly. It does not by itself identify $A_{\psi, \beta}(V)$ with a matrix algebra $M_q(\mathbb{C})$ — that identification exists (Theorem 3.6 of §3) but needs a polarization and is not canonical, whereas simplicity and the dimension count here need none.

Provenance. ids: D4, WH-COMM, WH-ALG; proved in: theory/wh-kappa.md §3; tested by: theory/checks/wh_kappa_check.py (C5, C6); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. Define the *Weyl average* $E : A_{\psi, \beta}(V) \rightarrow A_{\psi, \beta}(V)$ by

$$E(x) := q^{-2} \sum_{u \in V(\kappa)} W_\beta(u) x W_\beta(u)^{-1}.$$

Step 1: $E(\sum_v a_v W_\beta(v)) = a_0 \cdot 1$. By Theorem 2.7(d), $W_\beta(u)W_\beta(v)W_\beta(u)^{-1} = \psi(\omega(u, v))W_\beta(v)$, so

$$E\left(\sum_v a_v W_\beta(v)\right) = \sum_v a_v \left(q^{-2} \sum_u \psi(\omega(u, v))\right) W_\beta(v).$$

Fix $v \neq 0$. By Theorem 2.6(a), (c), $\omega(\cdot, v)$ is a nonzero κ -linear map $V(\kappa) \rightarrow \kappa$, hence surjective (a nonzero linear map between finite-dimensional vector spaces over the same field onto a 1-dimensional target is onto). So $u \mapsto \psi(\omega(u, v))$ is a character of the additive group $V(\kappa)$ that is nontrivial (since ψ is nontrivial and $\omega(\cdot, v)$ is onto κ). A nontrivial character χ of a finite abelian group G sums to zero: fixing u_0 with $\chi(u_0) \neq 1$, the substitution $u \mapsto u + u_0$ permutes G , so $S := \sum_u \chi(u)$ satisfies $S = \chi(u_0)S$, forcing $S = 0$ since $\chi(u_0) \neq 1$. Applying this to $\chi = \psi(\omega(\cdot, v))$ gives $\sum_u \psi(\omega(u, v)) = 0$ for $v \neq 0$; for $v = 0$ the sum is $q^2 \cdot \psi(0) = q^2$. Hence $E(\sum_v a_v W_\beta(v)) = a_0 \cdot W_\beta(0) = a_0 \cdot 1$, since $W_\beta(0)$ is the unit (Theorem 2.7(a)).

Step 2: $Z(A_{\psi,\beta}(V)) = \mathbb{C} \cdot 1$. If z is central then $W_\beta(u)zW_\beta(u)^{-1} = z$ for every u , so $z = E(z) = a_0 \cdot 1$ by Step 1; conversely every scalar is central.

Step 3: simplicity. Let I be a nonzero two-sided ideal and pick $0 \neq x = \sum_v a_v W_\beta(v) \in I$ with some coefficient $a_{v_0} \neq 0$. Then $y := W_\beta(v_0)^{-1}x \in I$ (as I is a left ideal), and by Theorem 2.7(b),(c) each $W_\beta(v_0)^{-1}W_\beta(v)$ is a nonzero scalar multiple of $W_\beta(v - v_0)$; in particular the coefficient of $W_\beta(0)$ in y is a nonzero scalar multiple of a_{v_0} , hence nonzero. Now $E(y) \in I$: E is a $\mathbb{C}$ -linear combination of the maps $x \mapsto W_\beta(u)xW_\beta(u)^{-1}$, and I is a two-sided ideal, so each term $W_\beta(u)yW_\beta(u)^{-1}$ lies in I and so does their average. By Step 1, $E(y) = c \cdot 1$ with $c \neq 0$ (the coefficient of $W_\beta(0)$ in y), so $1 \in I$ and $I = A_{\psi,\beta}(V)$.

Step 4: dimension. $\dim_{\mathbb{C}} A_{\psi,\beta}(V) = |V(\kappa)| = q^2$ directly from Definition 2.4. $\square$

Remark (Why the quantifier over β can be widened after the fact). Theorems 2.7 and 2.8 are stated for *every* $\beta \in \text{Adm}(\omega)$, not merely the reference cocycle β_0 . This widening is legitimate precisely because no step of either proof used more than the defining relation $\beta - \beta^T = \omega$ and bi-additivity: the commutator computation in Theorem 2.7(c)–(d) depends on β only through $\omega = \beta - \beta^T$, and simplicity in Theorem 2.8 depends only on ω being nondegenerate. That the choice of β is nonetheless *not* immaterial to everything built from it — it is invisible to the pair $(A_{\psi,\beta}(V), \text{Weyl frame})$ but not to the Heisenberg group $H_\beta(\kappa)$ itself once $p = 2$ — is the subject of §4.

3 Lagrangian lines, Schrödinger models, and the uniqueness of the finite Weyl representation

Recollection (the setting of §2). For a finite field $\kappa = \mathbb{F}_q$, $q = p^m$, the symplectic object is $V(\kappa) := \kappa \oplus \kappa$ with the κ -bilinear, alternating, nondegenerate form $\omega((a, b), (a', b')) := ab' - a'b$ (Definition 2.1); an admissible polarizing cocycle is a κ -bilinear $\beta : V(\kappa) \times V(\kappa) \rightarrow \kappa$ with $\beta - \beta^T = \omega$ (Definition 2.2); a phase datum is a nontrivial additive character $\psi : (\kappa, +) \rightarrow \mathbb{C}^\times$ (Definition 2.3); and the twisted algebra $A_{\psi,\beta}(V) := \bigoplus_{v \in V(\kappa)} \mathbb{C} \cdot W_\beta(v)$ has product $W_\beta(v)W_\beta(v') := \psi(\beta(v, v'))W_\beta(v + v')$ (Definition 2.4); by Theorem 2.8 it is simple of dimension q^2 with centre $\mathbb{C} \cdot 1$, for every $\beta \in \text{Adm}(\omega)$.

Provenance. ids: D1, D2, D3, D4; proved in: —; tested by: —; reviewed in: —.

This section names the Lagrangian lines of $V(\kappa)$, the finite-dimensional modules of $A_{\psi,\beta}(V)$ they induce, and proves that there is exactly one such module up to unitary equivalence and that it realizes $A_{\psi,\beta}(V)$ as a full matrix algebra — a finite-field analogue of the Stone–von Neumann theorem — together with a genuinely characteristic-two phenomenon in how the induced characters behave.

3.1 Preliminary facts about finite abelian characters and cocycles

Four short, elementary facts about characters of finite abelian groups recur throughout the rest of the labbook; they are recorded once, here, and cited by name afterward. None of them is part of the claims DAG — they carry no claim id and no status label, being standard and immediate from the definitions of “character” and “cocycle” — but each is proved in full, since nothing in this labbook is asserted from memory.

Fact 1 (orthogonality). *If G is a finite abelian group and $\chi : G \rightarrow \mathbb{C}^\times$ a nontrivial character, then $\sum_{u \in G} \chi(u) = 0$. Proof.* Pick $u_0 \in G$ with $\chi(u_0) \neq 1$. Then $S := \sum_u \chi(u) = \sum_u \chi(u + u_0) = \chi(u_0) S$, because $u \mapsto u + u_0$ permutes G ; since $\chi(u_0) \neq 1$, $S = 0$.

Fact 2 (valuation). *If ψ is a phase datum for κ (Definition 2.3) then $\psi(\kappa) \subseteq \mu_p$; $\ker \psi$ is an $\mathbb{F}_p$ -subspace of κ of index p ; and for $u \in \kappa$, $\psi(u) \equiv 1$ iff $u = 0$. Proof. $\psi(t)^p = \psi(pt) = \psi(0) = 1$ since $\text{char } \kappa = p$, so $\psi(\kappa) \subseteq \mu_p$. $\ker \psi$ is an additive subgroup of a characteristic- p field, hence an $\mathbb{F}_p$ -subspace, and $\kappa/\ker \psi$ embeds nontrivially into $\mu_p \cong \mathbb{Z}/p$ (Definition 2.3), forcing index exactly p . If $u \neq 0$ then $u\kappa = \kappa \not\subseteq \ker \psi$, i.e. $\psi(u) \neq 1$.*

Fact 3 (duality of elementary abelian groups). *Let U be a finite $\mathbb{F}_p$ -vector space of dimension d . Then $\hat{U} := \text{Hom}(U, \mathbb{C}^\times)$ has order p^d , and every element of $\hat{U}$ takes values in μ_p . Proof. For $\chi \in \hat{U}$, $\chi(u)^p = \chi(pu) = 1$, so $\chi(U) \subseteq \mu_p \cong \mathbb{Z}/p$, i.e. $\hat{U} = \text{Hom}_{\mathbb{F}_p}(U, \mu_p)$. Fixing an $\mathbb{F}_p$ -basis of U , a homomorphism is a free choice of d values in μ_p , so $|\hat{U}| = p^d$. (Applied below to $U = L$ a κ -line, an $\mathbb{F}_p$ -space of dimension $m = [\kappa : \mathbb{F}_p]$, giving $|\hat{L}| = p^m = q$; later to $U = \kappa$ and $U = V(\kappa)$, of $\mathbb{F}_p$ -dimension m and $2m$, giving $|\hat{\kappa}| = q$ and $|\hat{V}| = q^2$.)*

Fact 4 (triviality of symmetric cocycles of exponent p). *Let U be a finite abelian group of exponent p and $c : U \times U \rightarrow \mathbb{C}^\times$ a symmetric 2-cocycle: $c(u, u') = c(u', u)$ and $c(u, u')c(u + u', u'') = c(u, u' + u'')c(u', u'')$. Then there is $\mu : U \rightarrow \mathbb{C}^\times$ with $c(u, u') = \mu(u + u')/(\mu(u)\mu(u'))$ for all u, u' . Proof. On $E := \mathbb{C}^\times \times U$ define $(z, u)(z', u') := (zz'c(u, u'), u + u')$; the cocycle identity gives associativity and symmetry of c gives commutativity, so E is a finite abelian group. Induction inside E gives $(1, u)^p = (\gamma(u), 0)$ with $\gamma(u) := c(u, u)c(2u, u) \cdots c((p-1)u, u)$. Fix an $\mathbb{F}_p$ -basis $u_1, \dots, u_n$ of U ; as $\mathbb{C}$ is algebraically closed, choose z_i with $z_i^p = \gamma(u_i)^{-1}$ and put $f_i := (z_i, u_i)$, so $f_i^p = 1$. The subgroup $H := \langle f_1, \dots, f_n \rangle \leq E$ is then abelian of exponent dividing p , and since U is free on $\{u_i\}$ there is a unique $\mathbb{F}_p$ -linear $\sigma : U \rightarrow H$ with $\sigma(u_i) = f_i$ and $\text{pr}_U \circ \sigma = \text{id}_U$. Writing $\sigma(u) = (\mu(u), u)$, that σ is a homomorphism reads exactly $\mu(u)\mu(u')c(u, u') = \mu(u + u')$.*

3.2 Lagrangian lines and the algebra they generate

Definition 3.1 (The quadratic form of a polarizing cocycle). For $\beta \in \text{Adm}(\omega)$, define $Q_\beta : V(\kappa) \rightarrow \kappa$ by $Q_\beta(v) := \beta(v, v)$. For the reference cocycle, $Q_{\beta_0}(a, b) = ab$.

Scope. Under a symmetrized cocycle (available only at odd p ; §4) Q_β would vanish identically — that convention does not exist here, and Q_β is not zero. This definition asserts nothing beyond the formula: it does not by itself claim Q_β is a quadratic form in the technical sense (quadratic homogeneity and a bilinear polarization), which is established only where needed, at Theorem 4.5 of §4.

Provenance. ids: D6; proved in: —; tested by: —; reviewed in: —.

Definition 3.2 (Schrödinger models, and the standard model). For a κ -line $L \subset V(\kappa)$ put $A_L := \text{span}_{\mathbb{C}}\{W_\beta(l) : l \in L\} \subseteq A_{\psi, \beta}(V)$. Given a unital $\mathbb{C}$ -algebra homomorphism (character) $\chi : A_L \rightarrow \mathbb{C}$, the *Schrödinger model* of (L, χ) is

$$M_{L, \chi} := A_{\psi, \beta}(V) \otimes_{A_L} \mathbb{C}_\chi.$$

The *standard model* (taken for $\beta = \beta_0$) is $M_0 := \bigoplus_{y \in \kappa} \mathbb{C} \cdot e_y$, with $\{e_y\}$ declared orthonormal and

$$W(a, b)e_y := \psi(-b(y + a))e_{y+a}, \quad \text{i.e.} \quad W(a, b) = Z(-b)X(a),$$

where $X(a)e_y := e_{y+a}$ and $Z(b)e_y := \psi(by)e_y$. **This is a stipulation.**

Scope. The sign in $W(a, b) = Z(-b)X(a)$ is not cosmetic: the naive assignment $W(a, b) = Z(b)X(a)$, without the sign, realizes a *different* cocycle ($-ab'$, not β_0 's ab'), and the discrepancy

between the two is invisible at $p = 2$ (where $-1 = 1$) and visible only at odd p — a characteristic-dependent failure mode this definition is written to avoid by fixing the correct sign explicitly rather than leaving it to convention. That M_0 , so defined, genuinely satisfies the multiplication law of Definition 2.4 is verified as part of the proof of Theorem 3.5 below (Step 1) and checked computationally by gate C11 of `theory/checks/wh_kappa_check.py`.

Provenance. ids: D8; proved in: —; tested by: —; reviewed in: —.

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Theorem 3.3 (Lagrangian lines, and the Schrödinger models they carry). *Let κ , $V(\kappa)$, ω , $\beta \in \text{Adm}(\omega)$, ψ , $A_{\psi,\beta}(V)$, Q_β (Definition 3.1) and A_L , $M_{L,\chi}$ (Definition 3.2) be as above, for any finite field κ , $p = 2$ included. Then:*

- (a) *every κ -line $L \subset V(\kappa)$ is ω -isotropic ($\omega(v, v') = 0$ for all $v, v' \in L$), hence maximal isotropic; so Lagrangian line and line coincide as conditions on $L \subset V(\kappa)$, at every characteristic;*
- (b) *there are exactly $q + 1 = |\mathbb{P}^1(\kappa)|$ such lines;*
- (c) *A_L is commutative of dimension q , and carries exactly q characters, forming a torsor under $\hat{L} = \text{Hom}(L, \mathbb{C}^\times)$;*
- (d) $\dim_{\mathbb{C}} M_{L,\chi} = q$;
- (e) **(precision, binding on this row.)** *A Schrödinger model is a pair (L, χ) : there are $q(q + 1)$ such pairs, q per line — but by Theorem 3.5 below, all of them are isomorphic as $A_{\psi,\beta}(V)$ -modules. The count $q(q + 1)$ is therefore a count of labels, not of isomorphism classes of module. Among the $q + 1$ lines, only those with $Q_\beta|_L \equiv 0$ carry the trivial character $\chi \equiv 1$ (since $\chi \equiv 1$ is a character of A_L iff $Q_\beta(v_0)\kappa \subseteq \ker \psi$ iff $Q_\beta(v_0) = 0$, for $L = \kappa v_0$), and how many lines have $Q_\beta|_L \equiv 0$ is itself β -dependent: exactly 2 for the reference cocycle β_0 (§4 computes it in general).*

Scope. This theorem does not assert that distinct pairs (L, χ) give distinct, let alone non-isomorphic, modules — to the contrary, part (e) is precisely the observation that they do not, once Theorem 3.5 is available. It does not address the characteristic-two phenomenon in the values a character of A_L can take; that is Theorem 3.4 below. Nothing here depends on which $\beta \in \text{Adm}(\omega)$ was chosen beyond the formula for Q_β .

Provenance. ids: D1, D4, D6, D8, WH-FORM, WH-ALG, WH-POL; proved in: `theory/wh-kappa.md` §4; tested by: `theory/checks/wh_kappa_check.py` (C7); reviewed in: `theory/verdicts/wh-kappa-r1.md`.

Proof. (a) Write $L = \kappa v_0$. For $c, c' \in \kappa$, $\omega(cv_0, c'v_0) = cc' \omega(v_0, v_0) = 0$ by κ -bilinearity (Theorem 2.6(a)) and the computed alternating property (Theorem 2.6(b)) — at $p = 2$ antisymmetry alone would not give this, so the computed form of (b) is exactly what is used. Since $\dim_{\kappa} V(\kappa) = 2$ and ω is nondegenerate, no subspace of dimension 2 (i.e. all of $V(\kappa)$) is isotropic, so a line is a maximal isotropic subspace: *every* line is automatically Lagrangian, and the isotropy condition singles out nothing among lines.

(b) The nonzero vectors of $V(\kappa)$ number $q^2 - 1$, and each line contains $q - 1$ of them, so there are $(q^2 - 1)/(q - 1) = q + 1 = |\mathbb{P}^1(\kappa)|$ lines.

(c) By Definitions 2.4 and 3.1, $W(cv_0)W(c'v_0) = \psi(\beta(cv_0, c'v_0))W((c+c')v_0) = \psi(cc'Q_\beta(v_0))W((c+c')v_0)$, using $\beta(cv_0, c'v_0) = cc'\beta(v_0, v_0) = cc'Q_\beta(v_0)$ by κ -bilinearity. The phase is symmetric in (c, c') , so A_L is commutative; $\dim_{\mathbb{C}} A_L = |L| = q$ since $\{W(l) : l \in L\}$ is linearly independent (distinct basis elements of $A_{\psi,\beta}(V)$). *Characters exist:* $c(c, c') := \psi(cc'Q_\beta(v_0))$ is a symmetric bi-additive, hence symmetric 2-cocycle, on $(L, +)$, a group of exponent p ; Fact 3.1 supplies

$\mu : L \rightarrow \mathbb{C}^\times$ with $c(c, c') = \mu(c + c')/(\mu(c)\mu(c'))$, and $\chi(W(cv_0)) := \mu(c)^{-1}$ is then an algebra character of A_L (it converts the twisted product into the untwisted one). *Torsor*: for two characters χ, χ' of A_L , the map $l \mapsto \chi'(W(l))/\chi(W(l))$ is a homomorphism $L \rightarrow \mathbb{C}^\times$, and conversely composing a character with any element of $\hat{L}$ gives another character; the action is free (a homomorphism fixing every $\chi(W(l))$ is trivial), and $|\hat{L}| = q$ by Fact 3.1 ($U = L$, $d = m$).

(d) $A_{\psi, \beta}(V)$ is free as a right A_L -module on any set of coset representatives $v_1, \dots, v_q$ of L in $V(\kappa)$: the elements $\{W(v_i)W(l) : l \in L\}$ are, up to the nonzero scalars $\psi(\beta(v_i, l))$, exactly the basis elements $\{W(v_i + l) : l \in L\}$ of the coset $v_i + L$. So $A_{\psi, \beta}(V) \cong A_L^{\oplus q}$ as right A_L -modules, and $M_{L, \chi} = A_{\psi, \beta}(V) \otimes_{A_L} \mathbb{C}_\chi$ has dimension $q^2/q = q$.

(e) There are q characters per line by (c) and $q + 1$ lines by (b), giving $q(q + 1)$ pairs (L, χ) ; that they are all isomorphic as $A_{\psi, \beta}(V)$ -modules is Theorem 3.5(c) below (which uses only parts (a)–(d) of this theorem, not this part, so there is no circular dependency). For $L = \kappa v_0$, $\chi \equiv 1$ is a character of A_L iff $c(c, c') = \psi(cc'Q_\beta(v_0)) \equiv 1$ iff $Q_\beta(v_0)\kappa \subseteq \ker \psi$; since $Q_\beta(v_0)\kappa$ is either $\{0\}$ or all of κ (it is a κ -subspace, being the image of scalar multiplication by $Q_\beta(v_0)$) and $\ker \psi \neq \kappa$ (Fact 3.1), this holds iff $Q_\beta(v_0) = 0$. For $\beta = \beta_0$, $Q_{\beta_0}(a, b) = ab$ (Definition 3.1) vanishes exactly on the two coordinate lines $\{a = 0\}$ and $\{b = 0\}$, so exactly 2 of the $q + 1$ lines carry the trivial character. $\square$

3.3 The characteristic-two phase phenomenon

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Theorem 3.4 (Forced order-four phases in the induced characters, at characteristic two). *Let $L = \kappa v_0 \subset V(\kappa)$ be a line with $Q_\beta(v_0) \neq 0$ (Definition 3.1), and let χ be any character of A_L (Definition 3.2, Theorem 3.3(c)).*

- [$p = 2$.] χ takes a value of exact multiplicative order 4 on some $W(l)$, $l \in L$: the phases of the model are forced out of $\mu_2 = \psi(\kappa)$ into μ_4 . For the reference cocycle β_0 , the lines with $Q_{\beta_0}|_L \equiv 0$ are exactly the two coordinate axes (Theorem 3.3(e)), so this phenomenon applies to $q - 1$ of the $q + 1$ lines.
- [p odd.] No such phenomenon occurs: every character of A_L can be, and by the construction of Theorem 3.3(c) is, μ_p -valued.

Scope. This theorem says nothing about lines with $Q_\beta|_L \equiv 0$ (there the trivial character is available and no order-4 value is forced), and nothing about the algebra $A_{\psi, \beta}(V)$ globally — only about the values a character of the specific commutative subalgebra A_L can take. It does not identify *which* μ_4 this is inside a larger structure; that question, and its relation to the level- μ_p Weyl frame, is taken up in §4.

Provenance. ids: D6, D8, WH-POL, WH-POL-2; proved in: theory/wh-kappa.md §4 step (1)5, (1)6; tested by: theory/checks/beta_census.py (B6, $\#\{Q=0\}$); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. Write $Q_0 := Q_\beta(v_0) \neq 0$ and $\nu(c) := \chi(W(cv_0))$ for the character χ constructed in the proof of Theorem 3.3(c); $\nu(0) = 1$, and setting $c' = c$ in the cocycle relation there, $\nu(c)^2 = \psi(c^2Q_0)\nu(2c)$.

[$p = 2$.] Here $2c = 0$, so $\nu(2c) = \nu(0) = 1$ and $\nu(c)^2 = \psi(c^2Q_0)$. The Frobenius map $c \mapsto c^2$ is injective, hence bijective, on the finite field κ , so $\{c^2Q_0 : c \in \kappa\} = Q_0\kappa = \kappa$ (as $Q_0 \neq 0$). Since ψ is nontrivial, $\psi(c^2Q_0) = -1$ for some $c \in \kappa$ — because -1 is the unique nontrivial element of $\mu_2 = \{1, -1\}$ and $\psi(\kappa) = \mu_2$ by Fact 3.1 — giving $\nu(c)^2 = -1$, i.e. $\nu(c)$ has exact order 4. The count of lines follows from Theorem 3.3(e).

[p odd.] Put $\mu(c) := \psi(c^2Q_0/2)$ (division by 2 makes sense, p being odd). Then, directly, $\psi((c^2 + c'^2)Q_0/2) \psi(cc'Q_0) = \psi((c + c')^2Q_0/2)$, i.e. $\mu(c)\mu(c')\psi(cc'Q_0) = \mu(c + c')$: μ already solves the cocycle relation of Theorem 3.3(c) with values entirely inside $\psi(\kappa) = \mu_p$, so no order-4 (or any value outside μ_p) is ever forced. $\square$

3.4 Uniqueness of the finite Weyl representation

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Theorem 3.5 (Uniqueness of the finite Weyl representation). *Let $\kappa, \beta \in \text{Adm}(\omega)$, $\psi \in X(\kappa)$, and $A_{\psi,\beta}(V)$ be as above. Then:*

- (a) *the induced module M_0 of (L_0, χ_0) , $L_0 := 0 \oplus \kappa$, $\chi_0 \equiv 1$, has basis $\{e_y\}_{y \in \kappa}$ with $W_{\beta_0}(a, b)e_y = \psi(-b(y+a))e_{y+a}$ — i.e. the stipulated standard model of Definition 3.2 genuinely satisfies the multiplication law of Definition 2.4;*
- (b) *every simple $A_{\psi,\beta}(V)$ -module is unitarily equivalent to M_0 (equivalently, to any $M_{L,\chi}$), and an $A_{\psi,\beta}(V)$ -linear map $M_0 \rightarrow M_0$ is a scalar;*
- (c) *consequently $H_\beta(\kappa)$ has, up to unitary equivalence, exactly one irreducible unitary representation with central character ψ , of dimension q , and every $\mathbb{C}$ -algebra automorphism of $A_{\psi,\beta}(V)$ is inner;*
- (d) *for $\{e_y\}$ orthonormal, every $W_{\beta_0}(v)$ acts unitarily, and in general the unitary intertwiner between two irreducible unitary representations with the same central character is unique up to a scalar in $U(1)$.*

Scope. This theorem is stated and proved without invoking Maschke’s theorem, Artin–Wedderburn, Skolem–Noether, Jacobson density, or Pontryagin duality as black boxes: every step is elementary, using only Theorem 2.8 (simplicity) and Theorem 3.3. It is uniform in the characteristic: no clause is restricted to odd p or to $p = 2$, and no step below treats $p = 2$ differently. It does not exhibit the isomorphism $A_{\psi,\beta}(V) \cong M_q(\mathbb{C})$ as canonical — that non-canonicity is the separate content of Theorem 3.6 immediately below.

Provenance. ids: D1, D4, D8, WH-ALG, WH-POL, WH-SVN; proved in: theory/wh-kappa.md §5; tested by: theory/checks/wh_kappa_check.py (C3,C6,C11); reviewed in: theory/verdicts/wh-kappa-r1.md; theory/verdicts/wh-kappa-adjudication.md.

Proof. (a) *The standard model, verified.* Since $Q_{\beta_0}(0, b) = 0 \cdot b = 0$, $\chi_0 \equiv 1$ is a character of A_{L_0} by Theorem 3.3(e). Take coset representatives $(y, 0)$, $y \in \kappa$, of L_0 in $V(\kappa)$ and put $e_y := W_{\beta_0}(y, 0) \otimes 1 \in M_{L_0, \chi_0}$. Then

$$W_{\beta_0}(a, b) e_y = \psi(\beta_0((a, b), (y, 0))) W_{\beta_0}(a + y, b) \otimes 1 = W_{\beta_0}(a + y, b) \otimes 1,$$

since $\beta_0((a, b), (y, 0)) = a \cdot 0 = 0$. To evaluate this against χ_0 the L_0 -part must be moved to the right:

$$\begin{aligned} W_{\beta_0}(a + y, 0) W_{\beta_0}(0, b) &= \psi((a + y)b) W_{\beta_0}(a + y, b), \quad \text{so} \\ W_{\beta_0}(a + y, b) &= \psi(-b(y + a)) W_{\beta_0}(a + y, 0) W_{\beta_0}(0, b). \end{aligned}$$

Tensoring against $\chi_0(W_{\beta_0}(0, b)) = 1$ gives $W_{\beta_0}(a, b)e_y = \psi(-b(y + a))e_{a+y}$ as claimed, and $\dim M_0 = q$ by Theorem 3.3(d). (The sign here is exactly Definition 3.2’s stipulation, now derived rather than assumed: the naive un-signed assignment $W(a, b) = Z(b)X(a)$ realizes the *different* cocycle $-ab'$ and fails to satisfy Definition 2.4’s multiplication law for $\beta_0(v, v') = ab'$ at odd p — gate C11 of theory/checks/wh_kappa_check.py reports 0 violations for the signed model at all six tested $q \in \{2, 3, 4, 5, 8, 9\}$, against 36, 400, 3888 violations for the unsigned model at $q = 3, 5, 9$ and, tellingly, 0 violations at $q = 2, 4, 8$ — the defect is invisible at $p = 2$, where $-1 = 1$, and visible only once p is odd.)

(b) *Uniqueness and Schur.* $\pi : A_{\psi,\beta}(V) \rightarrow \text{End}_{\mathbb{C}}(M_0)$ is a unital, nonzero algebra homomorphism, so $\ker \pi$ is a proper two-sided ideal, hence 0 by simplicity (Theorem 2.8): π is injective. Since $\dim \text{End}_{\mathbb{C}}(M_0) = q^2 = \dim A_{\psi,\beta}(V)$, π is also surjective, giving $A_{\psi,\beta}(V) \cong \text{End}_{\mathbb{C}}(M_0)$.

Consequently $A_{\psi,\beta}(V) \cong M_0^{\oplus q}$ as a left module over itself (the columns of $\text{End}_{\mathbb{C}}(M_0)$), so *any* simple module N — generated by any nonzero $n \in N$ — is a quotient of $A_{\psi,\beta}(V) \cong M_0^{\oplus q}$; restricting the quotient map to one summand gives a nonzero map $M_0 \rightarrow N$, which is surjective (its image is a nonzero submodule of the simple N) and injective (its kernel is a proper submodule of the simple M_0), hence an isomorphism $M_0 \cong N$. For an A -endomorphism φ of M_0 : φ , as a $\mathbb{C}$ -linear endomorphism of a finite-dimensional $\mathbb{C}$ -vector space, has an eigenvalue $\lambda \in \mathbb{C}$ ($\mathbb{C}$ is algebraically closed), and $\ker(\varphi - \lambda)$ is then a nonzero A -submodule of the simple M_0 , so $\varphi = \lambda \cdot \text{id}$: $\text{End}_A(M_0) = \mathbb{C} \cdot \text{id}$.

(c) *Representations of $H_\beta(\kappa)$ and inner automorphisms.* A representation of $H_\beta(\kappa)$ with central character ψ is exactly an $A_{\psi,\beta}(V)$ -module (the remark after Theorem 2.7); irreducibility is simplicity of the module, so (b) applies: there is exactly one such representation up to isomorphism, of dimension q . For automorphisms: given $\alpha \in \text{Aut}_{\mathbb{C}}(A_{\psi,\beta}(V))$, twist the action on M_0 by α , i.e. put $a \cdot_\alpha m := \pi(\alpha(a))m$; this twisted module M_0^α is again simple of dimension q , hence $\cong M_0$ by (b), via some invertible T ; unwinding, $\alpha = \text{Ad}_T$, with T unique up to a scalar by the Schur clause of (b). No Skolem–Noether theorem is invoked.

(d) *Unitarity.* Since $|\psi(\cdot)| = 1$ (Fact 3.1), each $W_{\beta_0}(a, b)$ of (a) sends the orthonormal $\{e_y\}$ to $\{\psi(-b(y+a))e_{a+y}\}$, again orthonormal: every $W_{\beta_0}(v)$ is unitary for this inner product, with no averaging needed. (For a general β , the Weyl operators generate the *finite* level- μ_p frame $\mathcal{F}_{\psi,\beta}^{(p)}$ of Definition 4.9 in §4, of order pq^2 ; averaging any inner product on the simple module over this finite group makes every $\pi(W_\beta(v))$ unitary there too.) If T intertwines two unitary representations sharing central character ψ , then T^*T commutes with every operator of the first, so $T^*T = c \cdot \text{id}$ with $c > 0$ by the Schur clause of (b); then $c^{-1/2}T$ is unitary, and two unitary intertwiners differ by a unimodular scalar, i.e. an element of $U(1)$. $\square$

PROVED

Theorem 3.6 (The twisted algebra as a matrix algebra, non-canonically). *$A_{\psi,\beta}(V) \cong M_q(\mathbb{C})$, by an isomorphism that depends on a choice of Lagrangian line and character (L, χ) (Definition 3.2) and a choice of basis of $M_{L,\chi}$, and is **not** canonical.*

Scope. This theorem asserts existence of *an* isomorphism to $M_q(\mathbb{C})$ and the dependence of any such isomorphism on auxiliary choices; it does not assert, and Theorem 6.3 of §6 shows precisely how much of this data can be removed, that any two such isomorphisms agree, nor that a canonical one exists.

Provenance. ids: D4, D8, WH-ALG, WH-SVN, WH-ALG-MAT; proved in: theory/wh-kappa.md §5 step (1)2; tested by: theory/checks/wh-kappa-check.py (C5); reviewed in: theory/verdicts/wh-kappa-r1.md.

Proof. This is exactly the isomorphism $\pi : A_{\psi,\beta}(V) \xrightarrow{\sim} \text{End}_{\mathbb{C}}(M_0)$ constructed in the proof of Theorem 3.5(b); choosing the basis $\{e_y\}_{y \in \kappa}$ of M_0 identifies $\text{End}_{\mathbb{C}}(M_0)$ with $M_q(\mathbb{C})$. The construction visibly used a Lagrangian line L_0 , a character χ_0 of A_{L_0} , and the basis $\{e_y\}$ of the resulting module; a different line, character, or basis gives a different isomorphism, related to this one by an inner automorphism of $M_q(\mathbb{C})$ (Theorem 3.5(c)) composed with the change of basis, and no choice among these is distinguished by the data (κ, ψ, β) alone. $\square$

4 The polarizing cocycle as a second datum: the characteristic-two dichotomy

This section proves the increment’s central structural fact. The quantum system attached to a finite field κ is built not from the field alone, and not from the field together with a single choice of additive character ψ : it is built from *three* data — κ , ψ , and a polarizing cocycle β , a choice inside the torsor $\text{Adm}(\omega)$ of Definition 2.2 that is no more optional, and no more a matter of

convention, than ψ itself. What this section shows is exactly how much that third choice matters. At odd characteristic it turns out not to matter at all: a canonical representative exists inside $\text{Adm}(\omega)$, and every other admissible cocycle produces an isomorphic Heisenberg group by an isomorphism fixing the centre. At characteristic two no canonical representative exists, and the choice is not immaterial: the torsor $\text{Adm}(\omega)$ splits into exactly two orbits under the natural symmetry group $SL_2(\kappa)$, distinguished by the Arf invariant of an associated quadratic form, and the two orbits produce two genuinely non-isomorphic Heisenberg groups — a fact detectable by an operator identity inside the quantum system itself.

This is not a defect in the construction, and not a gap for a better convention to close. It is a theorem about what the construction actually depends on, carrying an intrinsic, computable invariant that says exactly when two choices of cocycle agree and when they do not. A sharp result of this shape — naming precisely the choice at stake and exhibiting the invariant that classifies it — is a positive structural statement about the definition, not a negative one, and it is this labbook’s principal new content.

4.1 The torsor of admissible polarizing cocycles, restated

Definition 4.1 (Admissible polarizing cocycles, restated in full). For the symplectic object $(V(\kappa), \omega)$ of Definition 2.1,

$$\text{Adm}(\omega) := \{ \beta : V(\kappa) \times V(\kappa) \rightarrow \kappa \mid \beta \text{ is } \kappa\text{-bilinear and } \beta - \beta^T = \omega \}.$$

A *polarizing cocycle* is a choice of element $\beta \in \text{Adm}(\omega)$, and it is a **datum** of the construction, on the same footing as the phase datum ψ — not a convention fixed once and for all. The *reference cocycle* is $\beta_0((a, b), (a', b')) := ab'$, a label for a point of the torsor below, not a canonical point of it. Write $\text{Sym}(V)$ for the set of symmetric κ -bilinear forms on $V(\kappa)$.

Scope. This restates Definition 2.2 verbatim; nothing new is stipulated here. What is new in this section is everything that is *proved* about $\text{Adm}(\omega)$: its torsor structure (Theorem 4.2), the immateriality of the choice at odd p (Theorem 4.3), and its genuine two-way split at $p = 2$ (Theorems 4.4–4.8).

Provenance. ids: D1, D2; proved in: —; tested by: —; reviewed in: —.

PROVED

Theorem 4.2 (The admissible cocycles form a torsor of size q^3). $\text{Adm}(\omega)$ is a torsor under $\text{Sym}(V)$ acting by $\beta \mapsto \beta + s$, $s \in \text{Sym}(V)$; consequently $|\text{Adm}(\omega)| = q^3$.

Scope. This is purely a statement about the set $\text{Adm}(\omega)$ and its free transitive action; it does not yet distinguish any point of the torsor, does not use the characteristic in any way, and does not address isomorphism of the associated Heisenberg groups — those are the later theorems of this section.

Provenance. ids: D1, D2, WH-BETA-a; proved in: theory/wh-kappa-choice.md §6 step ⟨1⟩1; tested by: theory/checks/beta_census.py (B1); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. If $\beta, \beta' \in \text{Adm}(\omega)$, put $s := \beta' - \beta$. Then $s - s^T = (\beta' - \beta^T) - (\beta - \beta^T) = \omega - \omega = 0$, i.e. $s \in \text{Sym}(V)$. Conversely, for $s \in \text{Sym}(V)$, $(\beta + s) - (\beta + s)^T = (\beta - \beta^T) + (s - s^T) = \omega + 0 = \omega$, so $\beta + s \in \text{Adm}(\omega)$. The action $\beta \mapsto \beta + s$ is therefore well defined, free (if $\beta + s = \beta + s'$ then $s = s'$), and transitive (any $\beta' \in \text{Adm}(\omega)$ equals $\beta + (\beta' - \beta)$ for the $s = \beta' - \beta \in \text{Sym}(V)$ just exhibited). A symmetric κ -bilinear form on $V(\kappa) = \kappa e \oplus \kappa f$ is determined freely by the three values $s(e, e), s(e, f), s(f, f) \in \kappa$ (symmetry forces $s(f, e) = s(e, f)$, and bilinearity determines everything else), so $|\text{Sym}(V)| = q^3$; since $\text{Adm}(\omega)$ is nonempty ($\beta_0 \in \text{Adm}(\omega)$) and a torsor under $\text{Sym}(V)$, it has the same cardinality, q^3 . $\square$

4.2 Odd characteristic: a canonical representative

PROVED [†]

Theorem 4.3 (Immateriality of the polarizing cocycle at odd characteristic). *[p odd.] $\text{Adm}(\omega)$ contains exactly one antisymmetric member, $\omega/2$, and it is $SL_2(\kappa)$ -invariant. For every $s \in \text{Sym}(V)$, the map*

$$\varphi_s(t, v) := \left(t + \frac{1}{2}s(v, v), v \right)$$

is a group isomorphism $H_\beta(\kappa) \rightarrow H_{\beta+s}(\kappa)$ fixing the centre $\kappa \times 0$ pointwise, for every $\beta \in \text{Adm}(\omega)$. Hence at odd characteristic the choice of polarizing cocycle is immaterial: every admissible cocycle produces a Heisenberg group isomorphic, by an explicit isomorphism fixing the centre, to every other, and $\omega/2$ is a canonical basepoint of the torsor.

Scope. This theorem is restricted to odd p by hypothesis, not by omission: at $p = 2$, division by 2 is division by zero, $\omega/2$ has no referent, and Theorem 4.4 shows the phenomenon this theorem describes simply does not occur. Nothing here compares $H_\beta(\kappa)$ for different β as algebras or as sources of quantum systems beyond the group isomorphism exhibited; the algebra-level statement is Theorem 4.10.

Provenance. ids: D2, D5, WH-BETA-a, WH-BETA-b; proved in: theory/wh-kappa-choice.md §6 step ⟨1⟩2; tested by: theory/checks/beta_census.py (B2,B3,B4); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Existence and uniqueness of the antisymmetric point. Since p is odd, $2 \in \kappa^\times$, and $\omega/2$ is κ -bilinear with $(\omega/2) - (\omega/2)^T = \omega/2 + \omega/2 = \omega$ (using $\omega^T = -\omega$, i.e. antisymmetry of ω , itself immediate from Theorem 2.6(b): $\omega(v, v') + \omega(v', v) = \omega(v + v', v + v') - \omega(v, v) - \omega(v', v') = 0$), so $\omega/2 \in \text{Adm}(\omega)$ and is manifestly antisymmetric. If $\beta, \beta' \in \text{Adm}(\omega)$ are both antisymmetric, then $s := \beta' - \beta \in \text{Sym}(V)$ (Theorem 4.2) is also antisymmetric, so $2s = 0$, so $s = 0$ (p odd): the antisymmetric point is unique. *Invariance.* $SL_2(\kappa)$ preserves ω (Theorem 2.6(e)), hence preserves the scalar multiple $\omega/2$.

The isomorphism φ_s . Put $f(v) := s(v, v)/2$. By bilinearity and symmetry of s , $s(v + v', v + v') = s(v, v) + 2s(v, v') + s(v', v')$, so $f(v + v') = f(v) + f(v') + s(v, v')$. Unwinding the group law of Definition 2.5 for $H_\beta(\kappa)$ and $H_{\beta+s}(\kappa)$,

$$\varphi_s((t, v)(t', v')) = \varphi_s(t + t' + \beta(v, v'), v + v') = (t + t' + \beta(v, v') + f(v + v'), v + v'),$$

while

$$\varphi_s(t, v) \varphi_s(t', v') = (t + f(v), v)(t' + f(v'), v') = (t + t' + f(v) + f(v') + (\beta+s)(v, v'), v + v'),$$

computed inside $H_{\beta+s}(\kappa)$. The two right-hand sides agree precisely because $f(v + v') = f(v) + f(v') + s(v, v')$. So φ_s is a group homomorphism; it is a bijection (identity on the second coordinate, an affine bijection $t \mapsto t + f(v)$ on the first, for each fixed v), hence an isomorphism, and $\varphi_s(t, 0) = (t + f(0), 0) = (t, 0)$ since $f(0) = s(0, 0)/2 = 0$: the centre is fixed pointwise. $\square$

4.3 Characteristic two: no symmetric distinguished point

PROVED [†]

Theorem 4.4 (Absence of a symmetric distinguished cocycle at characteristic two). *[$p = 2$.] $\text{Adm}(\omega)$ contains no antisymmetric member; the criterion that singles out a canonical basepoint at odd characteristic (Theorem 4.3) distinguishes nothing here.*

Scope. This theorem only rules out *this particular* route to a canonical choice (antisymmetry). It does not by itself assert that no other route to a canonical choice exists, nor that the resulting Heisenberg groups actually differ — those stronger, positive facts are Theorems 4.5–4.7 below.

Provenance. ids: D2, WH-BETA-a, WH-BETA-c; proved in: theory/wh-kappa-choice.md §6 step ⟨1⟩3; tested by: theory/checks/beta_census.py (B2); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. At $p = 2$, $-1 = 1$, so “antisymmetric” ($\beta = -\beta^T$) coincides with “symmetric” ($\beta = \beta^T$), i.e. $\beta - \beta^T = 0$. But every $\beta \in \text{Adm}(\omega)$ has $\beta - \beta^T = \omega$ (Definition 4.1), and $\omega((1, 0), (0, 1)) = 1 \neq 0$ (Theorem 2.6(c)), so $\omega \neq 0$: no member of $\text{Adm}(\omega)$ can be symmetric, hence none can be antisymmetric either. $\square$

4.4 The quadratic form of a cocycle, and the order profile of its Heisenberg group

PROVED [†]

Theorem 4.5 (The quadratic form of a polarizing cocycle, and the element-order profile of its Heisenberg group). *[$p = 2$.] Let $\beta \in \text{Adm}(\omega)$, with $Q_\beta(v) := \beta(v, v)$ as in Definition 3.1. Then:*

- (i) $Q_\beta(cv) = c^2 Q_\beta(v)$ for $c \in \kappa$, $v \in V(\kappa)$, and $Q_\beta(v + v') = Q_\beta(v) + Q_\beta(v') + \omega(v, v')$ for all v, v' : Q_β is a κ -quadratic form on $V(\kappa)$ with polar form ω ;
- (ii) in $H_\beta(\kappa)$, $(t, v)^2 = (Q_\beta(v), 0)$ for every (t, v) ; consequently, writing $n_0 := |Q_\beta^{-1}(0)|$ (which is always ≥ 1 , since $Q_\beta(0) = 0$), the multiset of element orders of $H_\beta(\kappa)$ is

$$\{ 1 : 1, \quad 2 : q n_0 - 1, \quad 4 : q^3 - q n_0 \};$$

- (iii) hence $\text{Adm}(\omega)$ meets at least two isomorphism classes of Heisenberg group: by direct enumeration at $q = 2$, 6 of the 8 admissible cocycles give the order profile $\{1:1, 2:5, 4:2\}$ ($n_0 = 3$) and the remaining 2 give $\{1:1, 2:1, 4:6\}$ ($n_0 = 1$), and these are different multisets, hence non-isomorphic groups.

Scope. Part (iii) establishes only that at least two isomorphism classes occur, by exhibiting two different order profiles at the single value $q = 2$; it does not claim that exactly two classes occur for every q , nor identify the classes by name (this lane computes order profiles, not group names — that two of the eight cocycles at $q = 2$ happen to give the quaternion group and the other six the dihedral group of order 8 is a fact the orchestrator’s independent census records, but it is not part of what this row proves). The full classification, valid at every $q = 2^m$ and carrying an intrinsic invariant, is Theorem 4.7 below, which depends on this one.

Provenance. ids: D2, D6, WH-BETA-a, WH-BETA-d; proved in: theory/wh-kappa-choice.md §6 step ⟨1⟩4; tested by: theory/checks/beta_census.py (B3,B6); theory/checks/wh_kappa_check.py (C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (i) By κ -bilinearity, $Q_\beta(cv) = \beta(cv, cv) = c^2 \beta(v, v) = c^2 Q_\beta(v)$. Expanding $Q_\beta(v + v') = \beta(v + v', v + v') = \beta(v, v) + \beta(v, v') + \beta(v', v) + \beta(v', v')$; at $p = 2$, addition and subtraction coincide, so $\beta(v, v') + \beta(v', v) = \beta(v, v') - \beta(v', v) = \omega(v, v')$ (Definition 4.1), giving $Q_\beta(v + v') = Q_\beta(v) + Q_\beta(v') + \omega(v, v')$.

(ii) In $H_\beta(\kappa)$, $(t, v)(t, v) = (2t + \beta(v, v), 2v) = (Q_\beta(v), 0)$ at $p = 2$, since $2t = 0$ and $2v = 0$. If $(t, v) = (0, 0)$ this is the identity (order 1). If $(t, v) \neq (0, 0)$ and $Q_\beta(v) = 0$, then $(t, v)^2 = (0, 0)$, so (t, v) has order exactly 2; the number of such elements is $q n_0 - 1$ (for each of the n_0 vectors v with $Q_\beta(v) = 0$, all q values of t occur, and the pair $(t, v) = (0, 0)$ is excluded). If $Q_\beta(v) \neq 0$, then $(t, v)^2 = (Q_\beta(v), 0) \neq (0, 0)$, but $(t, v)^4 = ((t, v)^2)^2 = (2Q_\beta(v), 0) = (0, 0)$, so (t, v) has order exactly 4; there are $q(q^2 - n_0) = q^3 - q n_0$ such elements ($q^2 - n_0$ choices of v , q choices of t each). These three counts sum to $1 + (q n_0 - 1) + (q^3 - q n_0) = q^3 = |H_\beta(\kappa)|$, as they must.

(iii) The order profile just derived is a group-isomorphism invariant (an isomorphism of groups preserves the multiset of element orders), so different profiles certify non-isomorphic

groups. At $q = 2$, exhaustive enumeration over all 8 members of $\text{Adm}(\omega)$ (`theory/checks/beta_census.py`, gates B3 and B6) computes n_0 for each and finds exactly the two values $n_0 = 3$ (for 6 of the 8 cocycles, giving profile $\{1:1, 2:5, 4:2\}$ by the formula of (ii)) and $n_0 = 1$ (for the remaining 2, giving $\{1:1, 2:1, 4:6\}$); these are distinct multisets, exhibiting two isomorphism classes. $\square$

4.5 The Arf classification

Definition 4.6 (The Artin–Schreier map, and the type of a polarizing cocycle). For κ of characteristic 2, define $\wp : \kappa \rightarrow \kappa$ by $\wp(x) := x^2 + x$. The *type* of $\beta \in \text{Adm}(\omega)$ is

$$\text{Arf}(Q_\beta) := Q_\beta(e) Q_\beta(f) \in \kappa / \wp(\kappa),$$

computed in any symplectic κ -basis (e, f) of $V(\kappa)$, i.e. a basis with $\omega(e, f) = 1$.

Scope. This is a stipulation, meaningful only at $p = 2$. That the value $\text{Arf}(Q_\beta)$ does not depend on the choice of symplectic basis, that $\kappa / \wp(\kappa)$ has exactly two elements, and that both values actually occur among the members of $\text{Adm}(\omega)$, are not part of the definition; they are exactly the content of Theorem 4.7 below, whose provenance names the argument DAG’s current row for this material (the single source’s own cross-reference for this content predates a later renaming in the DAG).

Provenance. ids: D10; proved in: —; tested by: —; reviewed in: —.

SKETCH

Theorem 4.7 (Classification of characteristic-two polarizing cocycles by the Arf invariant). *[$p = 2$.] Let $\beta \in \text{Adm}(\omega)$.*

- (i) $\wp$ is additive with $\ker \wp = \{0, 1\} = \mathbb{F}_2$, so $\kappa / \wp(\kappa)$ has exactly two elements;
- (ii) Q_β has a nonzero zero (some $v \neq 0$ with $Q_\beta(v) = 0$) if and only if $\text{Arf}(Q_\beta) = 0$; in that case $n_0 = |Q_\beta^{-1}(0)| = 2q - 1$, and otherwise $n_0 = 1$. Consequently $\text{Arf}(Q_\beta)$ does not depend on the choice of symplectic basis used to compute it, and is invariant under $SL_2(\kappa)$;
- (iii) call β hyperbolic if $\text{Arf}(Q_\beta) = 0$ and anisotropic otherwise. Two members of $\text{Adm}(\omega)$ are carried to each other by an $SL_2(\kappa)$ -isometry of their quadratic forms if and only if they have the same type; both types occur, with $q(q+1)/2$ hyperbolic and $q(q-1)/2$ anisotropic quadratic forms compatible with ω , and stabilizer orders $|O(Q_\beta) \cap SL_2(\kappa)| = 2(q-1)$ (hyperbolic) and $2(q+1)$ (anisotropic);
- (iv) $H_\beta(\kappa) \cong H_{\beta'}(\kappa)$ if and only if $\text{Arf}(Q_\beta) = \text{Arf}(Q_{\beta'})$. So $\text{Adm}(\omega)$ meets exactly two isomorphism classes of Heisenberg group, indexed by Arf .

Scope. This theorem is marked SKETCH for a procedural reason recorded in `theory/verdicts/wh-kappa-adjudication.md`: it was written in the repair wave that followed this increment’s single hostile-critic round and has not itself been through one, per this campaign’s capped review loop (`PRD.md`) — not because a specific gap is known. It does not extend beyond $p = 2$ (Theorem 4.3 covers odd p by an entirely different, and there, complete, argument), and it does not address whether the type can be chosen compatibly along field embeddings — that question is recorded as open in §8.

Provenance. ids: D6, D10, WH-BETA-d, WH-BETA-TYPE; proved in: `theory/wh-kappa-choice.md` §6 step $\langle 1 \rangle 4, \langle 1 \rangle 5$; tested by: `theory/checks/beta_census.py` (B5, B6, B7); `theory/checks/wh_kappa_check.py` (C10); reviewed in: `theory/verdicts/wh-kappa-adjudication.md`.

Proof. (i) $\wp(x+y) - \wp(x) - \wp(y) = (x+y)^2 - x^2 - y^2 = 2xy = 0$ at $p = 2$, so $\wp$ is additive (a group homomorphism $(\kappa, +) \rightarrow (\kappa, +)$). Its kernel consists of the roots of $x^2 + x = 0$, a degree-2 polynomial, so $|\ker \wp| \leq 2$; both $x = 0$ and $x = 1$ are roots, so $\ker \wp = \{0, 1\} = \mathbb{F}_2$ exactly. By rank-nullity for the $\mathbb{F}_2$ -linear map $\wp$ on the m -dimensional $\mathbb{F}_2$ -space κ , $|\wp(\kappa)| = q/2$, so $|\kappa/\wp(\kappa)| = 2$.

(ii) In a symplectic basis (e, f) , write $v = ae + bf$; by Theorem 4.5(i), $Q_\beta(ae + bf) = \alpha a^2 + ab + \delta b^2$ with $\alpha := Q_\beta(e)$, $\delta := Q_\beta(f)$ (using $\omega(e, f) = 1$ for the cross term). If $b = 0$: $Q_\beta = \alpha a^2$ vanishes for some $a \neq 0$ iff $\alpha = 0$, and then $\text{Arf} = \alpha\delta = 0 \in \wp(\kappa)$. If $b \neq 0$, rescale to $b = 1$: $\alpha a^2 + a + \delta = 0$; multiplying by α (when $\alpha \neq 0$) gives $(\alpha a)^2 + (\alpha a) = \alpha\delta$, i.e. $\wp(\alpha a) = \alpha\delta$, solvable for a iff $\alpha\delta \in \wp(\kappa)$; when $\alpha = 0$ the equation $a + \delta = 0$ always has the solution $a = \delta$. So in every case, Q_β has a nonzero zero iff $\alpha\delta \in \wp(\kappa)$, i.e. iff $\text{Arf}(Q_\beta) = 0 \in \kappa/\wp(\kappa)$. When $\text{Arf}(Q_\beta) = 0$, the zero set of Q_β is a union of κ -lines (by homogeneity $Q_\beta(cv) = c^2 Q_\beta(v)$ of Theorem 4.5(i)); the computation above produces exactly two such lines (the two roots of $\wp(\alpha a) = \alpha\delta$ when $\alpha \neq 0$, or the lines $b = 0$ and $a = \delta$ when $\alpha = 0$), giving $n_0 = 2(q-1)+1 = 2q-1$ nonzero-vector zeros plus the origin; when $\text{Arf}(Q_\beta) \neq 0$, $n_0 = 1$ (only the origin). Since $|\kappa/\wp(\kappa)| = 2$ by (i), the class $\alpha\delta \bmod \wp(\kappa)$ is itself a basis-free condition — “ Q_β is isotropic” — so $\text{Arf}(Q_\beta)$ is independent of the symplectic basis chosen to compute it, and $SL_2(\kappa)$ -invariant since an $SL_2(\kappa)$ -image of a symplectic basis is again symplectic (Theorem 2.6(e)).

(iii) *Normal forms.* If $\text{Arf}(Q_\beta) = 0$, pick $e \neq 0$ with $Q_\beta(e) = 0$ (by (ii)) and any f with $\omega(e, f) = 1$ (possible by nondegeneracy of ω); replacing f by $f + Q_\beta(f)e$ changes $Q_\beta(f)$ to $Q_\beta(f) + Q_\beta(f)^2 Q_\beta(e) + Q_\beta(f)\omega(f, e)$, which simplifies (using $Q_\beta(e) = 0$ and $\omega(f, e) = -\omega(e, f) = 1$ at $p = 2$) to 0, giving $Q_\beta(ae + bf) = ab$: the hyperbolic normal form. If $\text{Arf}(Q_\beta) \neq 0$ then $Q_\beta(v) \neq 0$ for every $v \neq 0$ (by (ii)); pick any $e \neq 0$, rescale by $\lambda = Q_\beta(e)^{-1/2}$ (Frobenius is bijective on κ , so square roots exist and are unique) to arrange $Q_\beta(e) = 1$, then choose f with $\omega(e, f) = 1$; replacing f by $f + ae$ changes $Q_\beta(f)$ by $\wp(a)$ (direct computation from Theorem 4.5(i)), so $Q_\beta(f)$ can be moved to any chosen representative of its class modulo $\wp(\kappa)$. Both normalizations send a symplectic basis to a symplectic basis, hence are realized by elements of $SL_2(\kappa)$ (Theorem 2.6(e)): two quadratic forms of the same type are $SL_2(\kappa)$ -isometric, and by (ii) two of different type cannot be (they have different n_0 , an isometry invariant). *Counting.* Over a fixed symplectic basis, the κ -quadratic forms with polar form ω are exactly the q^2 maps $\alpha a^2 + ab + \delta b^2$ (parametrized freely by $(\alpha, \delta) \in \kappa^2$, by Theorem 4.5(i)); among these, $\#\{\alpha\delta \in \wp(\kappa)\} = q + (q-1)(q/2) = q(q+1)/2$ are hyperbolic (the case $\alpha = 0$ contributes q choices of δ , and each of the $q-1$ nonzero α pairs with the $q/2$ elements of $\wp(\kappa)$) and the remaining $q(q-1)/2$ are anisotropic — both classes nonempty for every q . By orbit-stabilizer inside $SL_2(\kappa)$, of order $q(q^2-1)$, acting transitively within each type: stabilizer orders $q(q^2-1)/(q(q+1)/2) = 2(q-1)$ (hyperbolic) and $q(q^2-1)/(q(q-1)/2) = 2(q+1)$ (anisotropic). *Groups.* If $\text{Arf}(Q_\beta) = \text{Arf}(Q_{\beta'})$, the normal-form argument gives $Q_{\beta'} = Q_\beta \circ g$ for some $g \in SL_2(\kappa)$; then $r(v, v') := \beta(gv, gv') - \beta'(v, v')$ is symmetric with $r(v, v) = 0$ for all v (both sides equal $Q_{\beta'}(v)$ by construction of g), so, fixing an $\mathbb{F}_2$ -basis b_i of $V(\kappa)$ and writing $v = \sum c_i b_i$, $h(v) := \sum_{i < j} c_i c_j r(b_i, b_j)$ satisfies $h(v + v') - h(v) - h(v') = r(v, v')$ (a direct expansion using r symmetric and bilinear), and $(t, v) \mapsto (t + h(v), gv)$ is then a group isomorphism $H_{\beta'}(\kappa) \rightarrow H_\beta(\kappa)$ by the same computation as in the proof of Theorem 4.3. If instead $\text{Arf}(Q_\beta) \neq \text{Arf}(Q_{\beta'})$, the two groups have different values of n_0 (by (ii)), hence different element-order profiles (Theorem 4.5(ii)), so no group isomorphism of any kind exists between them. $\square$

Remark. Direct enumeration (`theory/checks/beta_census.py`, gates B5–B7) confirms every clause above at $q = 2, 4, 8$: Arf is constant on $SL_2(\kappa)$ -orbits and takes exactly the two values across all q^3 members of $\text{Adm}(\omega)$; $|O(Q_\beta) \cap SL_2(\kappa)| = 2, 6, 14$ (hyperbolic) and $6, 10, 18$ (anisotropic), matching $2(q \mp 1)$ exactly; the element-order profiles come in exactly two sizes, $(6, 2)$, $(40, 24)$, $(288, 224)$, at $q = 2, 4, 8$ respectively; and the isomorphism constructed in the proof of (iii) is verified on all pairs within the hyperbolic class.

4.6 The type, detected inside the quantum system

SKETCH

Theorem 4.8 (The Arf type is visible inside the quantum system). *[p = 2.] Let $\beta \in \text{Adm}(\omega)$ and ψ a phase datum. Then, inside $A_{\psi,\beta}(V)$:*

$$W_\beta(v)^2 = \psi(Q_\beta(v)) \cdot 1 \quad \text{for every } v \in V(\kappa),$$

and

$$q^{-1} \sum_{v \in V(\kappa)} W_\beta(v)^2 = \varepsilon \cdot 1, \quad \varepsilon = \begin{cases} +1 & \beta \text{ hyperbolic } (\text{Arf}(Q_\beta) = 0), \\ -1 & \beta \text{ anisotropic.} \end{cases}$$

Equivalently, identifying $\kappa/\wp(\kappa) \cong \mathbb{F}_2$ via the trace $\text{Tr}_{\kappa/\mathbb{F}_2}$ of Definition 2.3, $\varepsilon = (-1)^{\text{Tr}(\text{Arf}(Q_\beta))}$.

Scope. The identity above is an equation inside the algebra $A_{\psi,\beta}(V)$ itself, so it exhibits $\text{Arf}(Q_\beta)$ as an invariant not merely of the abstract group $H_\beta(\kappa)$ but of the level- μ_p Weyl frame $\mathcal{F}_{\psi,\beta}^{(p)}$ of Definition 4.9 below. As with Theorem 4.7, on which it depends, the SKETCH label reflects only that this material postdates the increment's single hostile-critic round, per `theory/verdicts/wh-kappa-adjudication.md`.

Provenance. ids: D3, D4, D6, D10, WH-BETA-EPS; proved in: `theory/wh-kappa-choice.md §6 step <1>6`; tested by: `theory/checks/beta_census.py (B6,B8)`; reviewed in: `theory/verdicts/wh-kappa-adjudication.md`.

Proof. The square identity. By Definition 2.4 and Definition 3.1, $W_\beta(v)^2 = \psi(\beta(v, v))W_\beta(2v) = \psi(Q_\beta(v))W_\beta(0) = \psi(Q_\beta(v)) \cdot 1$, using $2v = 0$ at $p = 2$.

The hyperbolic value. For the hyperbolic normal form $Q(a, b) = ab$ (Theorem 4.7(iii)), $\sum_{a,b \in \kappa} \psi(ab) = \sum_a \left(\sum_b \psi(ab) \right)$; for $a \neq 0$, $b \mapsto \psi(ab)$ is a nontrivial character of κ , so the inner sum vanishes by Fact 3.1; for $a = 0$ the inner sum is q . So the double sum is q , giving $\varepsilon = +1$ for this normal form. Since $\sum_v \psi(Q_\beta(v))$ is invariant under precomposing Q_β with any bijection of $V(\kappa)$, in particular under the $SL_2(\kappa)$ -isometries of Theorem 4.7(iii), $\varepsilon = +1$ for *every* hyperbolic β , not only the normal form.

The anisotropic value. Sum $\sum_v \psi(Q(v))$ over *all* q^2 quadratic forms $Q(a, b) = \alpha a^2 + ab + \delta b^2$ compatible with ω (Theorem 4.7(iii)) and exchange the order of summation:

$$\sum_{\alpha, \delta \in \kappa} \sum_{a, b \in \kappa} \psi(\alpha a^2 + ab + \delta b^2) = \sum_{a, b \in \kappa} \psi(ab) \left(\sum_{\alpha} \psi(\alpha a^2) \right) \left(\sum_{\delta} \psi(\delta b^2) \right).$$

By Fact 3.1, $\sum_{\alpha} \psi(\alpha a^2) = q$ if $a = 0$ (else the character $\alpha \mapsto \psi(\alpha a^2)$ is nontrivial, since squaring is injective on κ , and the sum vanishes); likewise for the δ -sum. So only the term $a = b = 0$ survives, and the total is $q^2 \cdot \psi(0) = q^2$. By Theorem 4.7(iii) there are $q(q+1)/2$ hyperbolic forms, each contributing q (just shown), and $q(q-1)/2$ anisotropic forms, each contributing the same value S (by isometry-invariance, exactly as for the hyperbolic case). So $\frac{q(q+1)}{2} \cdot q + \frac{q(q-1)}{2} \cdot S = q^2$, giving $S = \frac{q^2 - \frac{q^2(q+1)}{2}}{\frac{q(q-1)}{2}} = \frac{q^2(1 - \frac{q+1}{2})}{\frac{q(q-1)}{2}} = \frac{-q^2(q-1)/2}{q(q-1)/2} = -q$. So $\varepsilon = -1$ on the anisotropic class.

Reading ε through the trace. $\text{Tr}_{\kappa/\mathbb{F}_2}$ is additive and $\mathbb{F}_2$ -valued; $\text{Tr}(\wp(x)) = \text{Tr}(x^2) + \text{Tr}(x) = \text{Tr}(x) + \text{Tr}(x) = 0$ (using $\text{Tr}(x^2) = \text{Tr}(x)$, since Frobenius permutes the terms of the trace sum, and $2 \text{Tr}(x) = 0$ at $p = 2$), so Tr kills $\wp(\kappa)$; it is nonzero (e.g. on a normal-basis generator), hence onto $\mathbb{F}_2$, and $\dim_{\mathbb{F}_2} \wp(\kappa) = m - 1 = \dim_{\mathbb{F}_2} \ker \text{Tr}$ by the rank computation of Theorem 4.7(i), so $\wp(\kappa) = \ker \text{Tr}$ exactly: Tr descends to an isomorphism $\kappa/\wp(\kappa) \xrightarrow{\sim} \mathbb{F}_2$. Under this identification the hyperbolic class ($\text{Arf} = 0$) corresponds to $\text{Tr} = 0$ and $\varepsilon = +1 = (-1)^0$, and the anisotropic class to $\text{Tr} = 1$ and $\varepsilon = -1 = (-1)^1$. $\square$

4.7 The level statement: what the algebra forgets, and what it does not

Definition 4.9 (The level- μ_p frame).

$$\mathcal{F}_{\psi,\beta}^{(p)} := \{ \zeta^j W_\beta(v) : j \in \mathbb{Z}/p, v \in V(\kappa) \} \subset A_{\psi,\beta}(V)^\times,$$

the finite group of Weyl unitaries with phases in $\mu_p = \psi(\kappa)$ (Fact 3.1), for ζ a primitive p -th root of unity as in Definition 2.3.

Scope. $\mathcal{F}_{\psi,\beta}^{(p)}$ is recorded separately from the Weyl frame $\mathcal{F}_\beta := \{ \mathbb{C}^\times W_\beta(v) : v \in V(\kappa) \} \subset A_{\psi,\beta}(V)$ (the set of $\mathbb{C}^\times$ -lines through Weyl operators; studied in §5 below) because the two behave differently under a change of β , exactly as Theorem 4.10 makes precise: the pair $(A_{\psi,\beta}(V), \mathcal{F}_\beta)$ turns out not to depend on β at all, while $\mathcal{F}_{\psi,\beta}^{(p)}$ — equivalently the isomorphism class of $H_\beta(\kappa)$ itself — does, and only at $p = 2$.

Provenance. ids: D11; proved in: —; tested by: —; reviewed in: —.

SKETCH

Theorem 4.10 (The algebra and its Weyl frame do not depend on β ; the level- μ_p frame does, at $p = 2$). *For all $\beta, \beta' \in \text{Adm}(\omega)$ there is an algebra isomorphism $A_{\psi,\beta}(V) \rightarrow A_{\psi,\beta'}(V)$ of the form $W_\beta(v) \mapsto \mu(v)W_{\beta'}(v)$ for some $\mu : V(\kappa) \rightarrow \mathbb{C}^\times$ — a frame isomorphism, carrying $\mathcal{F}_\beta$ to $\mathcal{F}_{\beta'}$ — and this holds **at every characteristic**. Moreover:*

- μ can be chosen μ_p -valued if and only if $\psi \circ Q_\beta = \psi \circ Q_{\beta'}$;
- $[p = 2]$ μ can always be chosen μ_4 -valued, regardless of type.

Hence the pair $(A_{\psi,\beta}(V), \mathcal{F}_\beta)$ does not depend on β at all, while the level- μ_p frame $\mathcal{F}_{\psi,\beta}^{(p)}$ — equivalently the isomorphism class of $H_\beta(\kappa)$ (Theorem 4.7) — does depend on β , and does so only at $p = 2$.

Scope. This theorem does not contradict Theorems 4.4–4.8: it is a statement about the algebra $A_{\psi,\beta}(V)$ and its $\mathbb{C}^\times$ -frame $\mathcal{F}_\beta$, which turn out to be β -independent, while those theorems are about the finite group $H_\beta(\kappa)$ (equivalently, the level- μ_p frame), which is not. The distinction between the two frames is exactly what makes both statements true simultaneously. This row is SKETCH for the same procedural reason as Theorems 4.7–4.8, and the explicit μ_2/μ_4 -valued formulas at $p = 2$ are presented at the level of detail the source shard gives them, verified computationally rather than re-derived symbol-by-symbol here.

Provenance. ids: D2, D4, D11, WH-BETA-a, WH-BETA-d, WH-BETA-LEVEL; proved in: theory/wh-kappa-choice.md §6 step ⟨1⟩7; tested by: theory/checks/frame_mu.py (M1,M2,M3,M4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Put $s := \beta' - \beta \in \text{Sym}(V)$ (Theorem 4.2). Direct expansion of the multiplication law of Definition 2.4 shows $v \mapsto \mu(v)W_{\beta'}(v)$ is an algebra map exactly when

$$\frac{\mu(v)\mu(v')}{\mu(v+v')} = \psi(-s(v, v')) \quad (\dagger)$$

for all v, v' , and any such map automatically carries $\mathcal{F}_\beta$ to $\mathcal{F}_{\beta'}$, since it sends each line $\mathbb{C}^\times W_\beta(v)$ to the line $\mathbb{C}^\times W_{\beta'}(v)$. The right-hand side $\psi \circ (-s)$ is a symmetric 2-cocycle on $(V(\kappa), +)$, a group of exponent p (every element of the characteristic- p field κ , hence of $V(\kappa)$, has additive order dividing p), so Fact 3.1 supplies a solution μ of $(\dagger)$: **the frame isomorphism exists at every characteristic**, with no restriction on β, β' .

Necessity of the μ_p criterion. Normalize $\mu(0) = 1$ (forced by $(\dagger)$ at $v = v' = 0$, since $\mu(0)^2/\mu(0) = \mu(0) = \psi(0) = 1$). At $p = 2$, setting $v' = v$ in $(\dagger)$ gives $\mu(v)^2/\mu(0) = \mu(v)^2 = \psi(s(v, v)) = \psi(Q_{\beta'}(v) - Q_{\beta}(v))$. So μ is μ_2 -valued (i.e. $\mu(v)^2 = 1$ for every v) if and only if $\psi(Q_{\beta'}(v)) = \psi(Q_{\beta}(v))$ for every v , i.e. iff $\psi \circ Q_{\beta} = \psi \circ Q_{\beta'}$; at odd p the analogous computation with $2 \in \kappa^{\times}$ gives the same criterion for μ_p -valuedness in general.

Sufficiency, $p = 2$. Fix an $\mathbb{F}_2$ -basis $b_1, \dots, b_{2m}$ of $V(\kappa)$ and write $\psi(s(b_i, b_j)) = (-1)^{\sigma_{ij}}$, $\sigma_{ij} \in \{0, 1\}$, for the finitely many structure values of the symmetric form s on this basis (well defined since $\psi(\kappa) = \mu_2 = \{\pm 1\}$ at $p = 2$, Fact 3.1). If $\sigma_{ii} = 0$ for every i (the case $\psi \circ Q_{\beta} = \psi \circ Q_{\beta'}$ just characterized), the $\mathbb{F}_2$ -bilinear cochain $\ell(v) := \sum_{i < j} c_i c_j \sigma_{ij}$, for $v = \sum_i c_i b_i$, solves $(\dagger)$ with values in $\mu_2 = \{\pm 1\}$, by a direct check against the defining bilinear expansion of s . In general — without assuming $\sigma_{ii} = 0$ — the integer-valued cochain $m(v) := 2 \sum_{i < j} c_i c_j \sigma_{ij} + \sum_i c_i \sigma_{ii} \pmod{4}$ solves $(\dagger)$ with $\mu(v) := i^{m(v)} \in \mu_4$ ($i = \sqrt{-1}$): this is the explicit construction of **theory/wh-kappa-choice.md** §6, verified against all $|V(\kappa)|^2$ pairs (v, v') for every one of the q^3 members of $\text{Adm}(\omega)$ at $q = 2, 4, 8$ by gates M2 and M3 of **theory/checks/frame_mu.py** (which finds μ_4 -valued solutions always, and μ_2 -valued solutions for exactly the q pairs (β, β') with $\psi \circ Q_{\beta} = \psi \circ Q_{\beta'}$, matching the necessity criterion above exactly).

Sufficiency, odd p . $\mu(v) := \psi(s(v, v)/2)$ solves $(\dagger)$: since s is symmetric bilinear, $s(v, v) + s(v', v') - s(v + v', v + v') = -2s(v, v')$, so $\mu(v)\mu(v')/\mu(v + v') = \psi([s(v, v) + s(v', v') - s(v + v', v + v')]/2) = \psi(-s(v, v'))$, matching $(\dagger)$ exactly, with values entirely inside $\psi(\kappa) = \mu_p$: this reproves, independently of Theorem 4.3, that nothing leaves the level- μ_p frame once p is odd — a second, algebra-level confirmation that the choice of β is immaterial there.

Gate M1 of **theory/checks/frame_mu.py** additionally searches all 2^{q^2} candidate μ_2 -valued phase functions exhaustively at $q = 2$ and confirms solutions exist exactly for the pairs the necessity criterion predicts; gate M4 checks the odd- p formula at $q = 3, 5, 9$. $\square$

5 Symmetry of the Weyl frame: the Weil representation and its splitting problem

Recollection. $\kappa = \mathbb{F}_q$, $q = p^m$; $V(\kappa) = \kappa \oplus \kappa$ with $\omega((a, b), (a', b')) = ab' - a'b$ (Definition 2.1); $SL_2(\kappa)$ is the group of κ -linear automorphisms of $V(\kappa)$ preserving ω (Theorem 2.6(e)); a polarizing cocycle $\beta \in \text{Adm}(\omega)$ and phase datum ψ determine the algebra $A_{\psi, \beta}(V) = \bigoplus_v \mathbb{C} \cdot W_{\beta}(v)$ (Definitions 4.1, 2.3, 2.4), and $Q_{\beta}(v) := \beta(v, v)$ (Definition 3.1).

Provenance. ids: D1, D2, D3, D4, D6; proved in: —; tested by: —; reviewed in: —.

This section studies the symmetries of the Weyl frame itself: which permutations of the Weyl operators extend to algebra automorphisms, how large that symmetry group is, and whether $SL_2(\kappa)$ — the natural symmetry group of $(V(\kappa), \omega)$ — genuinely acts on the quantum system by honest algebra automorphisms (as opposed to merely projectively). The answer is uniform in one respect and characteristic-dependent in another: the group of frame automorphisms always surjects onto $SL_2(\kappa)$ with kernel the dual group $\hat{V}$, at every characteristic; but whether this extension *splits* — whether $SL_2(\kappa)$ itself, not just a projective shadow of it, acts by algebra automorphisms — is settled affirmatively at odd p and only partially resolved at $p = 2$, where it interacts with the Arf dichotomy of §4.

5.1 The Weyl frame and its two automorphism groups

Definition 5.1 (The Weyl frame, and its κ -linear automorphisms). The *Weyl frame* is

$$\mathcal{F} := \{ \mathbb{C}^{\times} \cdot W_{\beta}(v) : v \in V(\kappa) \} \subset A_{\psi, \beta}(V),$$

the set of $\mathbb{C}^\times$ -lines through Weyl operators. $\text{Aut}_{\mathcal{F}}(A)$ is the group of $\mathbb{C}$ -algebra automorphisms α of $A_{\psi,\beta}(V)$ with $\alpha(\mathcal{F}) = \mathcal{F}$. $\text{Aut}_{\mathcal{F}}^\kappa(A) \subseteq \text{Aut}_{\mathcal{F}}(A)$ is the subgroup consisting of those α whose induced permutation of $V(\kappa)$ (well defined since α permutes the lines of $\mathcal{F}$, which are indexed bijectively by $V(\kappa)$) is κ -linear.

Scope. The κ -linearity clause in the definition of $\text{Aut}_{\mathcal{F}}^\kappa(A)$ is **imposed data**: nothing in $A_{\psi,\beta}(V)$ or $\mathcal{F}$ as built from $(V, +)$ and the cocycle $\psi \circ \beta$ forces an automorphism's induced permutation of $V(\kappa)$ to respect scalar multiplication by κ rather than merely by $\mathbb{F}_p$. Precisely how much symmetry this clause discards is Theorem 7.1 of §7; nothing here addresses that question.

Provenance. ids: D7; proved in: —; tested by: —; reviewed in: —.

5.2 Exactness of the frame-automorphism extension, uniformly in the characteristic

Fix $\beta \in \text{Adm}(\omega)$. For $g \in SL_2(\kappa)$, write $s_g(v, v') := \beta(gv, gv') - \beta(v, v')$ and $Q_g(v) := s_g(v, v)$; note that $Q_g(v) = Q_\beta(gv) - Q_\beta(v)$.

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Theorem 5.2 (Exactness of the frame-automorphism extension). *At every characteristic, the sequence*

$$1 \longrightarrow \hat{V} \longrightarrow \text{Aut}_{\mathcal{F}}^\kappa(A_{\psi,\beta}(V)) \xrightarrow{\alpha \mapsto g} SL_2(\kappa) \longrightarrow 1$$

is exact, where $\hat{V} := \text{Hom}((V(\kappa), +), \mathbb{C}^\times)$; the kernel consists exactly of the conjugations $\text{Ad}_{W_\beta(u)}$, $u \in V(\kappa)$.

Scope. This theorem does not assert that the extension splits — that $SL_2(\kappa)$ itself embeds back into $\text{Aut}_{\mathcal{F}}^\kappa(A)$ as a subgroup — only that it is exact. Splitting is addressed separately, and with a genuinely different answer at odd and even characteristic, in Theorems 5.3 and 5.6 below. Nothing here concerns $\text{Aut}_{\mathcal{F}}(A)$ without the imposed κ -linearity clause; that is Theorem 7.1.

Provenance. ids: D1, D4, D7, WH-ALG, WH-COMM, WH-WEIL-a; proved in: theory/wh-kappa-choice.md §8 step ⟨1⟩2, ⟨1⟩3; tested by: theory/checks/split24.py (X2); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. s_g is symmetric bi-additive, and $s_{gh}(v, v') = s_h(v, v') + s_g(hv, hv')$. Symmetry: $s_g(v, v') - s_g(v', v) = \omega(gv, gv') - \omega(v, v') = 0$, since $g \in SL_2(\kappa)$ preserves ω (Theorem 2.6(e)); the telescoping identity is immediate by expanding $\beta(ghv, ghv') - \beta(v, v')$ through the intermediate term $\beta(hv, hv')$. (For $\beta = \beta_0$ and $g = \begin{pmatrix} r & s \\ t & u \end{pmatrix}$, direct expansion of $\beta_0(gv, gv') = rt aa' + ru ab' + st a'b + su bb'$ gives the explicit formula $s_g(v, v') = rt aa' + st(ab' + a'b) + su bb'$, using $ru - st = \det g = 1$.)

Description of $\text{Aut}_{\mathcal{F}}^\kappa(A)$. Frame preservation gives $\alpha(W_\beta(v)) = \lambda(v)W_\beta(gv)$ for some function $\lambda : V(\kappa) \rightarrow \mathbb{C}^\times$ and some permutation g of $V(\kappa)$, imposed to be κ -linear by definition of $\text{Aut}_{\mathcal{F}}^\kappa(A)$. Applying α to $W_\beta(v)W_\beta(v') = \psi(\beta(v, v'))W_\beta(v + v')$ and comparing basis elements forces g additive (automatic here, being κ -linear) and

$$\lambda(v)\lambda(v')\psi(s_g(v, v')) = \lambda(v + v') \quad (*)$$

for all v, v' . Applying α to the commutation relation of Theorem 2.7(c) and using $\omega(gv, gv') = \det(g)\omega(v, v')$ (Theorem 2.6(e)'s proof) gives $\psi((\det g - 1)\omega(v, v')) = 1$ for all v, v' ; since ω is onto κ (Theorem 2.6(a),(c)), Fact 3.1 forces $\det g = 1$, i.e. $g \in SL_2(\kappa)$. So $\alpha \mapsto g$ is a well-defined map $\text{Aut}_{\mathcal{F}}^\kappa(A) \rightarrow SL_2(\kappa)$, visibly a homomorphism (composition of automorphisms composes the induced permutations).

The kernel. If $g = \text{id}$, $(*)$ reads $\lambda(v)\lambda(v') = \lambda(v + v')$, i.e. $\lambda \in \hat{V}$, a group of order q^2 by Fact 3.1 ($U = V(\kappa)$, $d = 2m$). The map $u \mapsto \psi(\omega(u, \cdot))$, $V(\kappa) \rightarrow \hat{V}$, is injective (if

$\psi(\omega(u, \cdot)) \equiv 1$ then $\omega(u, \cdot) \equiv 0$ by Fact 3.1, so $u = 0$ by nondegeneracy, Theorem 2.6(c)), hence bijective ($|V(\kappa)| = q^2 = |\hat{V}|$); so every kernel element has $\lambda = \psi(\omega(u, \cdot))$ for a unique $u \in V(\kappa)$, and by Theorem 2.7(d) this is exactly $\text{Ad}_{W_\beta(u)}$.

Surjectivity, uniformly in p . For fixed $g \in SL_2(\kappa)$, $c(v, v') := \psi(s_g(v, v'))$ is a symmetric 2-cocycle on the exponent- p group $(V(\kappa), +)$ (symmetry of c from symmetry of s_g , just shown; the cocycle identity from bi-additivity of s_g), so Fact 3.1 supplies λ solving (*): every $g \in SL_2(\kappa)$ is hit. This argument is uniform in the characteristic; nothing distinguishes $p = 2$. $\square$

Remark (Every frame automorphism is projectively unitary, at every characteristic). Fix a model (M, π) as in Theorem 3.5. Taking absolute values in (*) makes $v \mapsto |\lambda(v)|$ a homomorphism from the finite group $V(\kappa)$ to the torsion-free group $(\mathbb{R}_{>0}, \times)$, hence trivial: every λ occurring in $\text{Aut}_{\mathcal{F}}^\kappa(A)$ is automatically unimodular. By Theorem 3.5(c),(d), every $\alpha \in \text{Aut}_{\mathcal{F}}^\kappa(A)$ is Ad_T for some T , unique up to a scalar; since α permutes the *unitary* Weyl operators (unimodular λ times unitary $\pi(W_\beta(gv))$), T may be chosen unitary and is then unique up to $U(1)$ (Theorem 3.5(d)). This gives an injective homomorphism $\text{Aut}_{\mathcal{F}}^\kappa(A) \rightarrow PU(M)$: a projective unitary representation of $\text{Aut}_{\mathcal{F}}^\kappa(A)$, at every characteristic. Composed with a section $SL_2(\kappa) \rightarrow \text{Aut}_{\mathcal{F}}^\kappa(A)$ of Theorem 5.2's surjection — should one exist as an honest group homomorphism, i.e. should the extension split — this yields an honest (non-projective) representation of $SL_2(\kappa)$ by algebra automorphisms, and hence, via Ad , a projective unitary representation of $SL_2(\kappa)$ itself on M . What is *not* true is that this extends to all of $\text{Aut}_{\mathbb{C}}(A) \cong PGL_q(\mathbb{C})$: at $q = 2$, $T = \text{diag}(2, 1)$ conjugates $\pi(W(1, 0)) = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ to $\begin{pmatrix} 0 & 2 \\ 1/2 & 0 \end{pmatrix}$, whose singular values 2, $1/2$ differ, so it is not a scalar times a unitary — the frame hypothesis in Definition 5.1 is exactly what is needed above.

5.3 Splitting at odd characteristic

PROVED [†]

Theorem 5.3 (The Weil representation of $SL_2(\kappa)$, at odd characteristic). *[p odd.] $\alpha_g(W_\beta(v)) := \psi(Q_g(v)/2) W_\beta(gv)$ defines a splitting $g \mapsto \alpha_g$ of the extension of Theorem 5.2: $SL_2(\kappa)$ acts on $A_{\psi, \beta}(V)$ by $\mathbb{C}$ -algebra automorphisms, with phases in μ_p . Composed with Remark 5.2, this exhibits a projective unitary representation of $SL_2(\kappa)$ on the model space of Theorem 3.5.*

Scope. This is an odd-characteristic statement only; Theorem 5.6 addresses whether an analogous honest (non-projective) action of $SL_2(\kappa)$ by algebra automorphisms exists at $p = 2$, and the answer there is open in general. Nothing here claims the splitting is unique, or canonical among splittings.

Provenance. ids: D4, D6, WH-WEIL-a, WH-SVN, WH-WEIL-b; proved in: theory/wh-kappa-choice.md §8 *step* $\langle 1 \rangle 4, \langle 1 \rangle 7$; tested by: theory/checks/split24.py (X4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Since s_g is symmetric bilinear, $Q_g(v + v') = Q_g(v) + Q_g(v') + 2s_g(v, v')$ (the same polarization identity as in Definition 3.1's use elsewhere), so

$$\psi(Q_g(v + v')/2) = \psi(Q_g(v)/2) \psi(Q_g(v')/2) \psi(s_g(v, v')),$$

which is exactly relation (*) of Theorem 5.2's proof for $\lambda(v) := \psi(Q_g(v)/2)$: α_g is a well-defined element of $\text{Aut}_{\mathcal{F}}^\kappa(A)$ over g . That $g \mapsto \alpha_g$ is a homomorphism follows from the telescoping identity $s_{gh}(v, v') = s_h(v, v') + s_g(hv, hv')$ of Theorem 5.2's proof, evaluated at $v = v'$ to give $Q_{gh}(v) = Q_h(v) + Q_g(hv)$, whence $\alpha_g(\alpha_h(W_\beta(v))) = \psi(Q_h(v)/2) \psi(Q_g(hv)/2) W_\beta(ghv) = \psi(Q_{gh}(v)/2) W_\beta(ghv) = \alpha_{gh}(W_\beta(v))$. This is an honest algebra automorphism, so $SL_2(\kappa)$ acts by algebra automorphisms with μ_p phases (division by 2 makes sense since p is odd); the projective unitary statement is Remark 5.2 applied to this section. $\square$

5.4 Characteristic two: the orthogonal subgroup and forced order-four phases

PROVED [†]

Theorem 5.4 (μ_2 -phases occur exactly over the orthogonal subgroup of Q_β). [$p = 2$.] Write $O(Q_\beta) := \{g \in SL_2(\kappa) : Q_\beta \circ g = Q_\beta\}$. A frame automorphism $\alpha_{g,\lambda}$ (in the notation $\alpha(W_\beta(v)) = \lambda(v)W_\beta(gv)$ of Theorem 5.2's proof) has some choice of λ landing entirely in $\psi(\kappa) = \mu_2$ if and only if $g \in O(Q_\beta)$ — a condition on the chosen β , not on the characteristic alone. For the reference cocycle β_0 , $O(Q_{\beta_0}) \cap SL_2(\kappa)$ consists exactly of the matrices $\text{diag}(r, r^{-1})$ and $\text{antidiag}(s, s^{-1})$, $r, s \in \kappa^\times$; it has order $2(q-1)$ (equal to 2, 6, 14 at $q = 2, 4, 8$), and is a **proper** subgroup of $SL_2(\kappa)$ for every q .

Scope. This theorem is about which g admit a μ_2 -valued lift, for a fixed β ; it says nothing yet about what happens for $g \notin O(Q_\beta)$ (Theorem 5.5), and the concrete description of $O(Q_{\beta_0})$ is specific to the reference cocycle — for a general β of hyperbolic type the order is the same, $2(q-1)$ (Theorem 4.7(iii)), but the concrete matrices differ with the normal-form basis chosen there.

Provenance. ids: D6, WH-WEIL-a, D7, WH-WEIL-c; proved in: theory/wh-kappa-choice.md §8 step (1)5; tested by: theory/checks/beta_census.py (B6); theory/checks/fp_symmetry.py (S3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Setting $v' = v$ in (*) at $p = 2$ ($2v = 0$) gives $\lambda(v)^2 = \psi(Q_g(v))$. So a solution λ of (*) is μ_2 -valued (i.e. $\lambda(v)^2 = 1$ for every v) iff $Q_g(V(\kappa)) \subseteq \ker \psi$, i.e. iff $\psi \circ Q_g \equiv 1$.

For every β , this is already the exact isometry condition. $Q_g(v) = Q_\beta(gv) - Q_\beta(v)$ is additive in v : expanding $Q_\beta(g(v+v')) - Q_\beta(v+v')$ via Definition 3.1's polarization identity on both terms, the ω -cross-terms cancel because g preserves ω (Theorem 2.6(e)), leaving $Q_g(v+v') = Q_g(v) + Q_g(v')$; also $Q_g(cv) = c^2 Q_g(v)$ since g is κ -linear. So, in a basis, $Q_g(a, b) = \alpha a^2 + \delta b^2$ for some $\alpha, \delta \in \kappa$ (no cross term). If $\psi \circ Q_g \equiv 1$ then $\psi(\alpha a^2) = 1$ for all a (setting $b = 0$); since $a \mapsto a^2$ is bijective on κ , this forces $\alpha = 0$ by Fact 3.1, and likewise $\delta = 0$: $Q_g \equiv 0$, i.e. $g \in O(Q_\beta)$. Conversely, if $g \in O(Q_\beta)$ then trivially $\psi \circ Q_g \equiv 1$, and moreover a μ_2 -valued λ solving (*) then exists: $\psi \circ s_g$ is a symmetric cocycle with trivial diagonal ($\psi(s_g(v, v)) = \psi(Q_g(v)) = 1$), exactly the case handled by the ℓ -construction in the proof of Theorem 4.10 (§4). So $\psi \circ Q_g \equiv 1$ iff $g \in O(Q_\beta)$ iff a μ_2 -valued lift exists.

The reference cocycle. By the explicit formula of Theorem 5.2's proof, $s_g(v, v') = rt a a' + st(ab' + a'b) + sub b'$ for β_0 , so $Q_g(a, b) = rt a^2 + sub b^2$ (the cross term drops at $p = 2$). Thus $g \in O(Q_{\beta_0})$ iff $rt = 0$ and $su = 0$. Combined with $\det g = ru - st = 1$: if $r = 0$ then $st = -1 = 1$ (so $s, t \neq 0$), and $su = 0$ forces $u = 0$, giving $g = \begin{pmatrix} 0 & s \\ s^{-1} & 0 \end{pmatrix}$ (using $t = s^{-1}$ from $st = 1$) — an antidiagonal matrix; if $t = 0$ then $ru = 1$ (so $r, u \neq 0$, $u = r^{-1}$), and $su = 0$ forces $s = 0$, giving $g = \text{diag}(r, r^{-1})$; the case $r = t = 0$ contradicts $\det g = 1$. So $O(Q_{\beta_0}) \cap SL_2(\kappa) = \{\text{diag}(r, r^{-1})\} \cup \{\text{antidiag}(s, s^{-1})\}$, of order $(q-1) + (q-1) = 2(q-1)$ (each family is parametrized bijectively by $\kappa^\times$). Since $Q_{\beta_0}(a, b) = ab$ is the hyperbolic normal form (Theorem 4.7(iii)), this matches the general formula $2(q-1)$ there. Properness: $2(q-1) < q(q^2-1) = |SL_2(\kappa)|$ for every $q \geq 2$, since $q(q+1) \geq 6 > 2$. $\square$

PROVED [†]

Theorem 5.5 (μ_4 is forced exactly when the orthogonal subgroup is proper). [$p = 2$.] If $g \notin O(Q_\beta)$, every frame automorphism over g carries a phase of exact multiplicative order 4. Consequently μ_4 is forced somewhere in $\text{Aut}_{\mathcal{F}}^\kappa(A)$ exactly when $O(Q_\beta) \cap SL_2(\kappa)$ is a proper subgroup of $SL_2(\kappa)$: this holds for β_0 at every q (Theorem 5.4), and for every $\beta \in \text{Adm}(\omega)$ at $q = 4, 8$, by direct enumeration. It does **not** hold in the single case $q = 2$ with β anisotropic, where $O(Q_\beta) \cap SL_2(\mathbb{F}_2) = SL_2(\mathbb{F}_2)$ (order $6 = |O(Q_\beta) \cap SL_2(\kappa)|$ from the anisotropic formula $2(q+1)$ of Theorem 4.7(iii) at $q = 2$) and the entire group $\text{Aut}_{\mathcal{F}}^\kappa(A)$ admits μ_2 -valued phases

throughout. So μ_4 is **not** a characteristic-two invariant of $(V(\kappa), \omega, \psi)$ alone: whether it is forced depends on the type of β .

Scope. This theorem uses only Theorem 5.4 and the comparison of $2(q \mp 1)$ against $|SL_2(\kappa)| = q(q^2 - 1)$; it does not itself re-derive the general- q order $2(q + 1)$ for the anisotropic stabilizer (that is Theorem 4.7(iii), only cited here). The claim for *every* β at $q = 4, 8$ rests on direct enumeration rather than a closed-form argument for general q .

Provenance. ids: D6, D10, WH-WEIL-c, WH-WEIL-d; proved in: theory/wh-kappa-choice.md §8 step (1)6; tested by: theory/checks/split24.py (X4,X6, and the anisotropic cross-run at $q = 2$); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. By Theorem 5.4's proof, any λ solving $(*)$ over g satisfies $\lambda(v)^2 = \psi(Q_g(v))$ for every v . If $g \notin O(Q_\beta)$ then $Q_g \neq 0$, and (by the same bijective-squaring argument as there) $\psi(Q_g(v)) = -1$ for some v — since $Q_g(v) = -1$'s preimage under ψ being nontrivial requires only that $\psi(\kappa) = \mu_2 = \{1, -1\}$ (Fact 3.1) and Q_g nonzero, hence onto a nonzero multiple of κ , hits every value including one where $\psi = -1$. Then $\lambda(v)^2 = -1$, i.e. $\lambda(v)$ has exact order 4: *every* λ solving $(*)$ over this g (not just one choice) is forced to take an order-4 value at this particular v , since the equation $\lambda(v)^2 = \psi(Q_g(v)) = -1$ has no solution in μ_2 . Comparing $2(q - 1)$ and $2(q + 1)$ (Theorem 4.7(iii)) against $|SL_2(\kappa)| = q(q^2 - 1) = q(q - 1)(q + 1)$: $2(q - 1) < q(q - 1)(q + 1)$ always (as $q(q + 1) \geq 6$ for $q \geq 2$), so the hyperbolic stabilizer — in particular $O(Q_{\beta_0}) \cap SL_2(\kappa)$ — is always proper, giving some $g \notin O(Q_\beta)$ and hence a forced order-4 phase, at every q . For the anisotropic stabilizer, $2(q + 1) < q(q - 1)(q + 1)$ iff $2 < q(q - 1)$: this holds for $q \geq 4$ but *fails* at $q = 2$, where $q(q - 1) = 2$ exactly, so $2(q + 1) = 6 = q(q - 1)(q + 1) = |SL_2(\mathbb{F}_2)|$: the anisotropic stabiliser is the whole group at $q = 2$, so *no* g lies outside $O(Q_\beta)$, so no order-4 phase is forced by this argument, and — since μ_2 -valued lifts exist for every $g \in O(Q_\beta)$ by Theorem 5.4's proof, and $\hat{V}$ is automatically μ_2 -valued (Fact 3.1, exponent $p = 2$) — the whole of $\text{Aut}_{\mathcal{F}}^\kappa(A)$ admits μ_2 -valued phases in this one case. At $q = 4, 8$, exhaustive enumeration over every $\beta \in \text{Adm}(\omega)$ (theory/checks/split24.py, gates X4 and X6) confirms $O(Q_\beta) \cap SL_2(\kappa)$ is proper for both types, hence μ_4 is forced for every β there. $\square$

5.5 Does the extension split at characteristic two? A conjecture, and what is known

CONJECTURE

Conjecture 5.6 (Existence of a Weil representation of $SL_2(\kappa)$ at characteristic two). [$p = 2$.] *The extension of Theorem 5.2 splits for every $q = 2^m$: $SL_2(\kappa)$ acts on $A_{\psi, \beta}(V)$ by $\mathbb{C}$ -algebra automorphisms, for some choice of splitting. This is verified for the reference cocycle β_0 at $q = 2$ (exhaustive search finds exactly 4 distinct complements of $\hat{V}$ of order 6, over all generating pairs) and at $q = 4$ (64 lifts of a $(2, 3, 5)$ -presentation of $SL_2(\mathbb{F}_4)$ satisfy the defining relations and generate a subgroup of order 60 meeting $\hat{V}$ trivially); in both verified cases every complement necessarily uses phases of exact order 4 (no complement avoids μ_4 , since $|\hat{V}| \cdot |O(Q_{\beta_0}) \cap SL_2(\kappa)|$ does not divide $|\text{Aut}_{\mathcal{F}}^\kappa(A)|$'s μ_2 -part suitably: $6 \nmid 8$ at $q = 2$, $60 \nmid 96$ at $q = 4$). No argument is known for general q .*

Scope. This is recorded as a conjecture, not a theorem, precisely because the evidence is two finite instances and no general argument: per this campaign's discipline, a passing computational instance promotes nothing. The verified cases are for the reference cocycle β_0 only; splitting for other cocycles, and any relationship between splittings for cocycles of different Arf type, is not addressed. Should a splitting be found for general q , Remark 5.2 would then give a projective unitary representation of $SL_2(\kappa)$ at $p = 2$ exactly as Theorem 5.3 does at odd p .

Provenance. ids: D7, WH-WEIL-a, WH-WEIL-d, WH-WEIL-e; proved in: —; tested by: theory/checks/split24.py (X4,X5,X6); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

What is known, stated as evidence and not as a proof. At $q = 2$, $|\text{Aut}_{\mathcal{F}}^{\kappa}(A)| = q^2 |SL_2(\kappa)| = 4 \cdot 6 = 24$; exhaustive search over generating pairs of order-6 subgroups meeting $\hat{V} = \mathbb{Z}/2 \times \mathbb{Z}/2$ trivially finds exactly 4 complements, all using a phase of exact order 4 somewhere (`split24.py` gate X4). At $q = 4$, lifting a $(2, 3, 5)$ generator-relation presentation of $SL_2(\mathbb{F}_4) \cong A_5$ over all $16 \times 16 = 256$ candidate phase choices on the two generators finds exactly 64 lifts satisfying $x^2 = y^3 = (xy)^5 = 1$ and generating a subgroup of order 60 meeting $\hat{V}$ trivially, again always with order-4 phases (gate X5). That no complement can avoid μ_4 in either case is confirmed by comparing group orders directly (gate X6): the μ_2 -phase part of $\text{Aut}_{\mathcal{F}}^{\kappa}(A)$ has order $q^2 \cdot |O(Q_{\beta_0}) \cap SL_2(\kappa)| = 4 \cdot 2 = 8$ at $q = 2$ and $16 \cdot 6 = 96$ at $q = 4$, and $|SL_2(\kappa)| = 6$, respectively 60, does not divide either. This settles the existence question affirmatively at these two finite values and settles that any splitting there must use order-4 phases; it establishes nothing for $q = 2^m$, $m \geq 3$.

6 The choices the construction depends on, and what is canonical

Recollection. $\kappa = \mathbb{F}_q$; $\psi \in X(\kappa)$ a phase datum, $\beta \in \text{Adm}(\omega)$ a polarizing cocycle (Definitions 2.3, 4.1); $A_{\psi, \beta}(V)$ the resulting twisted algebra, simple of dimension q^2 (Theorem 2.8), with a unique simple unitary module up to unitary equivalence (Theorem 3.5); by Theorem 4.10, for fixed ψ every two choices of β give algebras related by an explicit frame isomorphism.

Provenance. ids: D2, D3, D4, WH-ALG, WH-SVN; proved in: —; tested by: —; reviewed in: —.

Sections 4 and 5 named the choices the construction depends on and measured exactly how much they matter. This section assembles that information into a single statement: precisely which data beyond κ the construction needs, and precisely what survives canonically once those choices are made. The headline fact is that the *projective* Hilbert space is canonical — it does not depend on which unitary model, which polarization, or (granting one more ingredient from §4) which polarizing cocycle was used to build it — even though the Hilbert space itself is not.

Definition 6.1 (The model groupoid, and its projectivization). $\text{Mod}_{\psi, \beta}(\kappa)$ is the category whose objects are pairs (M, π) with $\pi : A_{\psi, \beta}(V) \rightarrow \text{End}_{\mathbb{C}}(M)$ a unital algebra homomorphism making M a simple module and every $\pi(W_{\beta}(v))$ unitary, and whose morphisms $(M, \pi) \rightarrow (M', \pi')$ are unitary intertwiners $T : M \rightarrow M'$ ($T\pi(a) = \pi'(a)T$ for all a). $\text{PMod}_{\psi, \beta}(\kappa)$ is the same category with each Hom-set replaced by $\text{Hom}/U(1)$ (intertwiners identified when they differ by a unimodular scalar).

Scope. This definition alone asserts nothing about the size or connectivity of $\text{Mod}_{\psi, \beta}(\kappa)$; that it is nonempty, connected, and has automorphism group $U(1)$ at every object is Theorem 6.3(c) below, using Theorem 3.5, not part of the stipulation here.

Provenance. ids: D9; proved in: —; tested by: —; reviewed in: —.

6.1 Exactly two data beyond the field

PROVED

Theorem 6.2 (The construction depends on exactly two data beyond κ). (a) $X(\kappa) = \hat{\kappa} \setminus \{1\}$ has $q - 1$ elements and is a free transitive $\kappa^{\times}$ -torsor under $(u \cdot \psi)(x) := \psi(ux)$ (a single point when $q = 2$, where $\kappa^{\times}$ is trivial);

- (b) beyond κ , the construction of $A_{\psi,\beta}(V)$ uses **exactly two** further data: a nontrivial character $\psi \in X(\kappa)$ and a polarizing cocycle $\beta \in \text{Adm}(\omega)$ — given (ψ, β) , $A_{\psi,\beta}(V)$ (Definition 2.4) involves no further choice, while exhibiting a concrete Hilbert space (rather than the abstract algebra) needs a pair (L, χ) in addition (Definition 3.2);
- (c) distinct $\psi, \psi' \in X(\kappa)$ give representations of the same group $H_\beta(\kappa)$ (which, by Definition 2.5, does not involve ψ at all) with distinct central characters, hence inequivalent as representations of $H_\beta(\kappa)$;
- (d) $A_{\psi,\beta}(V) \cong A_{\psi',\beta'}(V)$ for every $\psi, \psi' \in X(\kappa)$ and $\beta, \beta' \in \text{Adm}(\omega)$; but the frame-preserving isomorphisms between two such algebras form a torsor under $\text{Aut}_{\mathcal{F}}^\kappa(A_{\psi,\beta}(V))$ (Theorem 5.2), a group of order $q^2 |SL_2(\kappa)| \geq 24$ — so no single isomorphism among them is distinguished by the data $(\kappa, \psi, \psi', \beta, \beta')$ alone.

Scope. Part (b) corrects an earlier working hypothesis of this campaign, that the construction depended on a single datum beyond κ : that hypothesis is refuted by Theorem 4.2 and Theorem 4.5 of §4, which show β is a second, independent choice. Part (d) asserts the algebras are abstractly isomorphic; it does not assert any *particular* isomorphism among them is preferred, and Theorem 6.3 below is precisely about what survives this non-uniqueness.

Provenance. ids: D2, D3, D4, WH-ALG, WH-BETA-a, WH-WEIL-a, WH-CHOICE; proved in: theory/wh-kappa-choice.md §7; tested by: theory/checks/wh_kappa_check.py (C8 in its E5 form, C10); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (a) $|\hat{\kappa}| = q$ by Fact 3.1 ($U = \kappa$, $d = m$), so $|X(\kappa)| = q - 1$. The action $(u \cdot \psi)(x) := \psi(ux)$ preserves nontriviality: if $\psi \not\equiv 1$ and $u \neq 0$ then, as $x \mapsto ux$ is a bijection of κ , $u \cdot \psi \equiv 1$ would force $\psi \equiv 1$, false. It is free: $u \cdot \psi = \psi$ means $\psi((u - 1)x) = 1$ for all x , i.e. $(u - 1)\kappa \subseteq \ker \psi$; if $u \neq 1$ then $(u - 1) \in \kappa^\times$ so $(u - 1)\kappa = \kappa \not\subseteq \ker \psi$ (Fact 3.1, ψ nontrivial), forcing $u = 1$. A free action of the order- $(q - 1)$ group $\kappa^\times$ on the order- $(q - 1)$ set $X(\kappa)$ has exactly one orbit of size $q - 1$, hence is transitive. At $q = 2$, $\kappa^\times = \{1\}$ and $X(\kappa)$ is a single point — the $q = 2$ choice, per Theorem 4.4, lives entirely in β , not in ψ .

(b) Immediate from Definitions 2.4 and 3.2: the structure constants of $A_{\psi,\beta}(V)$ are $\psi(\beta(v, v'))$, a function of κ, ψ, β alone, while $M_{L,\chi}$ additionally names a line L and a character χ of A_L .

(c) A representation of $H_\beta(\kappa)$ with central character ψ restricts on the centre $\kappa \times 0$ to $\psi \cdot \text{id}$ (the remark after Theorem 2.7); an intertwiner between representations with central characters $\psi \neq \psi'$ would have to commute with the centre acting as $\psi \cdot \text{id}$ on one side and $\psi' \cdot \text{id}$ on the other, forcing it to be 0 on the difference $(\psi(t) - \psi'(t)) \cdot \text{id} \neq 0$ for suitable t — no nonzero intertwiner exists, so the two representations of the fixed group $H_\beta(\kappa)$ are inequivalent.

(d) *Existence, for every pair.* By Theorem 4.10, $A_{\psi,\beta}(V) \cong A_{\psi,\beta_0}(V)$ for every β (via a frame isomorphism, taking $\beta' = \beta_0$ there), and symmetrically $A_{\psi',\beta'}(V) \cong A_{\psi',\beta_0}(V)$. For the reference cocycle, $W_\psi(a, b) \mapsto W_{\psi_u}(u^{-1}a, b)$ (for $\psi' = u \cdot \psi = \psi_u$ as in (a)) is an isomorphism $A_{\psi,\beta_0}(V) \rightarrow A_{\psi_u,\beta_0}(V)$, since $\beta_0(u^{-1}a, b) = u^{-1}\beta_0(a, b)$ makes the structure constants match: $\psi_u(\beta_0(u^{-1}a, u^{-1}b)) = \psi(u \cdot u^{-2}\beta_0(a, b)) = \psi(u^{-1}\beta_0(a, b))$, consistently with the images $W_{\psi_u}(u^{-1}a)$. Composing the three maps $(A_{\psi,\beta} \rightarrow A_{\psi,\beta_0} \rightarrow A_{\psi',\beta_0} \rightarrow A_{\psi',\beta'})$, the middle map since $X(\kappa)$ is a $\kappa^\times$ -torsor by (a)) gives an isomorphism $A_{\psi,\beta}(V) \rightarrow A_{\psi',\beta'}(V)$ for arbitrary $\psi, \psi', \beta, \beta'$.

Non-uniqueness. The frame-preserving isomorphisms between two fixed such algebras, once nonempty, form a free transitive right $\text{Aut}_{\mathcal{F}}^\kappa(A_{\psi,\beta}(V))$ -set under postcomposition (compose one such isomorphism with an automorphism of the source), and $|\text{Aut}_{\mathcal{F}}^\kappa(A_{\psi,\beta}(V))| = q^2 |SL_2(\kappa)|$ by Theorem 5.2, at least $4 \cdot 6 = 24$. More than one isomorphism exists whenever this group is non-trivial (always), and the data $(\kappa, \psi, \psi', \beta, \beta')$ do not name a preferred one: e.g. a family $\{h_u\}$, $h_u := \text{diag}(u^{-j}, u^{j-1})$ for a further arbitrary choice of $j \in \mathbb{Z}/(q - 1)$, gives infinitely many mutually inconsistent *coherent* systems of isomorphisms across the $\kappa^\times$ -action of (a) — the family of isomorphisms is trivializable, but not canonically: choosing one is a section, not a theorem. $\square$

6.2 The projective Hilbert space is canonical

PROVED

Theorem 6.3 (Canonicity of the projective Hilbert space, at fixed (κ, ψ, β)). (a) $\text{Mod}_{\psi, \beta}(\kappa)$ (Definition 6.1) is a nonempty, connected groupoid, with automorphism group $U(1)$ at every object;

(b) consequently every Hom -set of $\text{PMod}_{\psi, \beta}(\kappa)$ is a singleton, so every diagram in it commutes trivially, and the limit $\varprojlim_{\text{PMod}_{\psi, \beta}(\kappa)} P(M)$ exists with every projection an isomorphism; define $P(H_\psi(\kappa))$ to be this limit. The underlying Hilbert space $H_\psi(\kappa)$ itself is canonical only up to a $U(1)$ of identifications (one for each object of $\text{Mod}_{\psi, \beta}(\kappa)$ chosen as a representative), but the projective Hilbert space $P(H_\psi(\kappa))$ is canonical in (κ, ψ) , for fixed β .

Scope. This theorem fixes β throughout; that the resulting $P(H_\psi(\kappa))$ does not, in fact, depend on the choice of β either is the separate, weaker-status content of Theorem 6.4 below, which is not needed for and does not feed back into this proof. Nothing here addresses canonicity of a specific *vector* in $H_\psi(\kappa)$, only of the isomorphism class of the pair (M, π) up to the $U(1)$ ambiguity of unitary intertwiners.

Provenance. ids: D9, WH-SVN, WH-CHOICE, WH-CANON; proved in: theory/wh-kappa-choice.md §7 step ⟨1⟩3, ⟨1⟩4; tested by: —; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (a) Nonempty: Theorem 3.5(a) exhibits the standard model M_0 , unitary by Theorem 3.5(d). Connected: any two objects (M, π) , (M', π') are unitarily isomorphic by Theorem 3.5(b)–(d) (uniqueness of the simple module up to unitary equivalence), so a morphism exists between any two objects, in either direction. $\text{Aut}(M, \pi) = U(1)$: by Theorem 3.5(b)’s Schur clause, any self-intertwiner is a scalar; a unitary one is then a scalar of modulus 1, and every element of $U(1)$ is realized (scalars commute with everything, hence intertwine trivially).

(b) In a connected groupoid where every object has automorphism group $U(1)$, every Hom -set is a $U(1)$ -torsor (fix one isomorphism between two objects; every other differs from it by an automorphism of the source, i.e. by an element of $U(1)$). Quotienting each Hom -set by this $U(1)$ therefore collapses it to a single point: every Hom -set of $\text{PMod}_{\psi, \beta}(\kappa)$ is a singleton. Consequently, for any two morphisms f, g in $\text{PMod}_{\psi, \beta}(\kappa)$ with the same source and target, $f = g$ automatically (both are the unique element of that singleton Hom -set): every square of transition maps commutes, not by verification, but because there is only one morphism available to fill it. A category in which every Hom -set is a singleton is equivalent to the terminal (one-object, one-morphism) category, so the limit of the identity diagram $P(M) \rightsquigarrow P(M)$ over $\text{PMod}_{\psi, \beta}(\kappa)$ exists and every object, with its unique family of identifying isomorphisms to every other, computes it: this is $P(H_\psi(\kappa))$. Since the ordinary (non-projectivized) Hom -sets of $\text{Mod}_{\psi, \beta}(\kappa)$ are $U(1)$ -torsors and not singletons, no single Hilbert space among the objects is preferred over $U(1)$ -many identifications with it — but every one of them projects isomorphically onto the same limit $P(H_\psi(\kappa))$. $\square$

6.3 Independence from the polarizing cocycle as well

SKETCH

Theorem 6.4 (The projective Hilbert space does not depend on β either). $P(H_\psi(\kappa))$, as constructed in Theorem 6.3, is independent of the choice of polarizing cocycle $\beta \in \text{Adm}(\omega)$ used to build it: the frame isomorphism $A_{\psi, \beta}(V) \rightarrow A_{\psi, \beta'}(V)$ of Theorem 4.10 transports unitary simple modules and induces an equivalence $\text{Mod}_{\psi, \beta}(\kappa) \rightarrow \text{Mod}_{\psi, \beta'}(\kappa)$, which in turn induces the unique isomorphism between the two limits $P(H_\psi(\kappa))$ constructed at β and at β' .

Scope. This theorem is SKETCH for the same procedural reason as the $p = 2$ rows of §4 it depends on (Theorem 4.10): it postdates this increment's single hostile-critic round. It asserts that the *limit* is independent of β , not that any specific object of $\text{Mod}_{\psi,\beta}(\kappa)$ equals, on the nose, an object of $\text{Mod}_{\psi,\beta'}(\kappa)$ — the identification goes through the transport functor described in the proof, consistently with the fact that the underlying group $H_\beta(\kappa)$ genuinely changes isomorphism type with β at $p = 2$ (Theorem 4.7).

Provenance. ids: WH-CANON, WH-BETA-LEVEL, WH-CANON-BETA; proved in: theory/wh-kappa-choice.md §7 *step* ⟨1⟩4.⟨2⟩1; tested by: —; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. By Theorem 4.10 there is a frame isomorphism $\Phi : A_{\psi,\beta}(V) \rightarrow A_{\psi,\beta'}(V)$, $W_\beta(v) \mapsto \mu(v)W_{\beta'}(v)$, with μ unimodular (by the same argument as Remark 5.2: taking absolute values in the defining relation (†) of Theorem 4.10's proof makes $|\mu|$ a homomorphism from a finite group to $\mathbb{R}_{>0}$, hence trivial). Transport of structure along Φ sends an object (M, π) of $\text{Mod}_{\psi,\beta}(\kappa)$ to (M, π') with $\pi'(a') := \pi(\Phi^{-1}(a'))$ for $a' \in A_{\psi,\beta'}(V)$: M stays simple (an equivalence of module categories along an algebra isomorphism preserves simplicity), and each $\pi'(W_{\beta'}(v)) = \pi(\Phi^{-1}(W_{\beta'}(v))) = \mu(v)\pi(W_\beta(v))$ remains unitary (a unimodular scalar times a unitary operator). This is an equivalence of categories $\text{Mod}_{\psi,\beta}(\kappa) \rightarrow \text{Mod}_{\psi,\beta'}(\kappa)$, compatible with the projectivization functors to $\text{PMod}_{\psi,\beta}(\kappa)$ and $\text{PMod}_{\psi,\beta'}(\kappa)$ of Theorem 6.3(b). Both projectivized categories have singleton Hom-sets (Theorem 6.3(b), applied at β and at β' respectively), so the induced map between their limits $P(H_\psi(\kappa))_\beta \rightarrow P(H_\psi(\kappa))_{\beta'}$ is again the unique morphism available between two terminal-equivalent categories, hence an isomorphism, and the unique one compatible with the transport functor. $\square$

7 What κ -linear symmetry hides: the $\mathbb{F}_p$ -shadow of the form

Recollection. For $\beta \in \text{Adm}(\omega)$ and phase datum ψ , the Weyl frame is $\mathcal{F} = \{\mathbb{C}^\times W_\beta(v) : v \in V(\kappa)\} \subset A_{\psi,\beta}(V)$, and $\text{Aut}_{\mathcal{F}}(A)$ is the group of $\mathbb{C}$ -algebra automorphisms of $A_{\psi,\beta}(V)$ preserving $\mathcal{F}$ (Definition 5.1); $\text{Aut}_{\mathcal{F}}^\kappa(A) \subseteq \text{Aut}_{\mathcal{F}}(A)$ is the further subgroup whose induced permutation of $V(\kappa)$ is κ -linear — data *imposed* on top of $\text{Aut}_{\mathcal{F}}(A)$, not derived from it (Definition 5.1's scope). By Theorem 5.2, $\text{Aut}_{\mathcal{F}}^\kappa(A)$ surjects onto $SL_2(\kappa)$ with kernel $\hat{V}$, at every characteristic.

Provenance. ids: D4, D7, WH-WEIL-a; proved in: —; tested by: —; reviewed in: —.

This section makes good on the promissory note left in Definition 5.1: exactly how much symmetry the imposed κ -linearity clause discards. The answer is that $A_{\psi,\beta}(V)$ and its frame $\mathcal{F}$ do not, in fact, remember the κ -module structure of $V(\kappa)$ at all — they are built from the bare abelian group $(V(\kappa), +)$ and the cocycle $\psi \circ \beta$ — so the *full* frame symmetry group is larger than $SL_2(\kappa)$ whenever κ is a proper extension of its prime field, and the discarded symmetry is measured exactly.

PROVED

Theorem 7.1 (The twisted algebra does not remember the κ -structure of $V(\kappa)$). (a) $A_{\psi,\beta}(V)$ and its frame $\mathcal{F}$ are determined by the abelian group $(V(\kappa), +)$ together with the $\mathbb{C}^\times$ -valued function $\psi \circ \beta$ on $V(\kappa) \times V(\kappa)$ alone; nothing in either object refers to scalar multiplication by κ ;

(b) consequently, dropping the imposed κ -linearity clause of Definition 5.1, the frame automorphisms with merely $\mathbb{F}_p$ -linear induced permutation g of $V(\kappa)$ surject, with the same kernel $\hat{V}$, onto the isometry group $Sp(V(\kappa), \psi \circ \omega) \cong Sp_{2m}(\mathbb{F}_p)$ of the $\mathbb{F}_p$ -valued alternating form $\psi \circ \omega$; so $|\text{Aut}_{\mathcal{F}}(A)| = q^2 \cdot |Sp_{2m}(\mathbb{F}_p)|$;

- (c) $SL_2(\kappa) \subseteq Sp_{2m}(\mathbb{F}_p)$, with index exactly 1 when $m = 1$ and strictly greater than 1 as soon as $m > 1$: by complete enumeration, $|SL_2(\kappa)| = 6, 24, 60, 120, 504, 720$ against $|Sp_{2m}(\mathbb{F}_p)| = 6, 24, 720, 120, 1451520, 51840$ at $q = 2, 3, 4, 5, 8, 9$ respectively — indices 1, 1, 12, 1, 2880, 72. So Definition 5.1's κ -linearity clause discards genuine symmetry the algebra has, whenever κ is a proper extension of its prime field.

Scope. This theorem does not claim $Sp_{2m}(\mathbb{F}_p)$ acts by automorphisms preserving anything beyond the frame $\mathcal{F}$ and the abelian group structure — in particular it says nothing about whether this larger symmetry is compatible with the κ -linear structure used elsewhere in this labbook (Theorem 2.6(f) shows, to the contrary, that the κ -valued form ω itself sees only $SL_2(\kappa)$ among $\mathbb{F}_p$ -linear maps: the enlargement here is possible only because $\text{Aut}_{\mathcal{F}}(A)$ is built from the strictly weaker $\mathbb{F}_p$ -valued shadow $\psi \circ \omega$). At the prime field, $m = 1$, the two groups coincide and nothing is discarded; this is the campaign's guiding path.

Provenance. ids: D1, D7, WH-WEIL-a, WH-FORM, WH-SYMM; proved in: theory/wh-kappa-choice.md §9; tested by: theory/checks/fp_symmetry.py (S1,S2,S3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. (a) By Definition 2.4, the structure constants of $A_{\psi,\beta}(V)$ in the basis $\{W_\beta(v)\}$ are exactly the values $\psi(\beta(v, v'))$, indexed by pairs of elements of the abelian group $(V(\kappa), +)$; by Definition 5.1, $\mathcal{F}$ is the set of $\mathbb{C}^\times$ -lines through this same basis, again indexed only by $(V(\kappa), +)$. Neither construction refers to multiplication by elements of κ .

(b) Repeat the argument of Theorem 5.2's proof *without* imposing κ -linearity on g : frame preservation still gives $\alpha(W_\beta(v)) = \lambda(v)W_\beta(gv)$ with g an (a priori only additive) bijection of $V(\kappa)$ and λ satisfying relation $(*)$ there, since that derivation used only additivity of g , never κ -linearity. Applying α to Theorem 2.7(c), an identity entirely in $\psi \circ \omega$, gives $\psi(\omega(gv, gv')) = \psi(\omega(v, v'))$ for all v, v' — i.e. $g \in Sp(V(\kappa), \psi \circ \omega) := \{g \text{ additive} : \psi \circ \omega \circ (g \times g) = \psi \circ \omega\}$ — rather than the stronger $\omega \circ g = \omega$ used before. The surjectivity argument via Fact 3.1 applies verbatim to any such g (it never used κ -linearity, only that $(V(\kappa), +)$ has exponent p), so every $g \in Sp(V(\kappa), \psi \circ \omega)$ lifts; the kernel computation ($g = \text{id}$ forces $\lambda \in \hat{V}$, identified with $V(\kappa)$ via $u \mapsto \psi(\omega(u, \cdot))$) is unchanged, giving the same kernel $\hat{V}$ of order q^2 . Nondegeneracy of $\psi \circ \omega$ as an $\mathbb{F}_p$ -bilinear form on the $2m$ -dimensional $\mathbb{F}_p$ -space $V(\kappa)$: if $\psi(\omega(v, \cdot)) \equiv 1$ then $\omega(v, \cdot) \equiv 0$ (Fact 3.1, since $\omega(v, \cdot)$ is either 0 or onto κ , being κ -linear, and $\kappa \not\subseteq \ker \psi$), so $v = 0$ by Theorem 2.6(c). A nondegenerate alternating form on a $2m$ -dimensional space over $\mathbb{F}_p$ has isometry group (by definition, matching the notation notation.md fixes for $Sp_{2m}(\mathbb{F}_p)$) exactly $Sp_{2m}(\mathbb{F}_p)$.

(c) If $g \in SL_2(\kappa)$ then $\omega \circ (g \times g) = \omega$ (Theorem 2.6(e)), so certainly $\psi \circ \omega \circ (g \times g) = \psi \circ \omega$, i.e. $SL_2(\kappa) \subseteq Sp(V(\kappa), \psi \circ \omega)$. This containment is exactly the gap between Theorem 2.6(f) — which shows the $\mathbb{F}_p$ -linear isometries of the κ -valued ω are no larger than $SL_2(\kappa)$ — and the present, strictly weaker condition of isometry for the $\mathbb{F}_p$ -valued shadow $\psi \circ \omega$ alone: an $\mathbb{F}_p$ -linear g can preserve $\psi \circ \omega$ without preserving ω itself. Complete enumeration (theory/checks/fp_symmetry.py, gates S1 and S2) computes both orders directly at $q = 2, 3, 4, 5, 8, 9$, matching the closed form $|Sp_{2m}(\mathbb{F}_p)| = p^{m^2} \prod_{i=1}^m (p^{2i} - 1)$ in every case, and confirms the indices 1, 1, 12, 1, 2880, 72; at $m = 1$ (the prime field) the two group orders coincide, since $Sp_2(\mathbb{F}_p) = SL_2(\mathbb{F}_p)$ by definition in dimension 2, and nothing is discarded. $\square$

Remark (The characteristic-two orthogonal phenomenon survives the enlargement). The μ_2 -phase criterion of Theorem 5.4 ($g \in O(Q_\beta)$) makes sense verbatim for merely $\mathbb{F}_p$ -linear g , at $p = 2$: $\psi \circ Q_\beta \circ g = \psi \circ Q_\beta$ is a condition on an $\mathbb{F}_p$ -valued quadratic form $\psi \circ Q_\beta$ with polar form $\psi \circ \omega$, so the relevant subgroup inside the enlarged symmetry group of Theorem 7.1 is $O(\psi \circ Q_\beta) \cap Sp_{2m}(\mathbb{F}_p)$. Direct enumeration (gate S3) finds its order to be 2, 72, 40320 at $q = 2, 4, 8$, of index 3, 10, 36 in $Sp_{2m}(\mathbb{F}_p)$ — exactly the *same* index that $|O(Q_\beta) \cap SL_2(\kappa)| = 2, 6, 14$ has inside $SL_2(\kappa)$ (Theorem 5.4). So the characteristic-two orthogonal phenomenon of §5 is not an

artefact of the imposed κ -linearity in Definition 5.1: it survives, at the same relative index, inside the larger, intrinsic symmetry group $Sp_{2m}(\mathbb{F}_p)$ that Theorem 7.1 exhibits. This observation is recorded as part of the same source theorem as Theorem 7.1, under the same provenance (`theory/wh-kappa-choice.md` §9, step $\langle 1 \rangle 4$).

Remark (Reading). Two independent non-canoncities are now on record, and they are logically independent of each other. Theorem 4.7 of §4 shows the system does not remember which polarizing cocycle β was chosen, in a way that is invisible at odd characteristic and genuinely two-valued at $p = 2$. This theorem shows the system does not remember the κ -module structure of $V(\kappa)$ itself, at every characteristic and every $m > 1$, regardless of β . Both are statements about what the definition actually depends on, in the sense of `PRD.md`’s “canonical” requirement: any assignment $\text{Spec } R \mapsto \text{quantum system}$ general enough to restrict to this one on $\text{Spec } \mathbb{F}_q$ carries both phenomena, not as defects, but as exactly the symmetry and the choice the construction is built from.

8 Functoriality in the field: what extends along embeddings, and what does not

Recollection. A phase datum for $\kappa = \mathbb{F}_{p^m}$ is a nontrivial additive character $\psi : (\kappa, +) \rightarrow \mathbb{C}^\times$, and the trace-normalized family is $\psi_\zeta(z) := \zeta^{\text{Tr}_{\kappa/\mathbb{F}_p}(z)}$ for ζ a primitive p -th root of unity and $\text{Tr}_{\kappa/\mathbb{F}_p}(z) := z + z^p + \cdots + z^{p^{m-1}}$ (Definition 2.3); by Theorem 6.2(a), $X(\kappa)$ is a $\kappa^\times$ -torsor, so every $\psi \in X(\kappa)$ is $\psi_c := \psi_\zeta(c \cdot)$ for a unique $c \in \kappa^\times$.

Provenance. ids: D3; proved in: —; tested by: —; reviewed in: —.

Earlier sections fixed a single finite field κ throughout. This section asks how the construction behaves as κ varies along field embeddings: whether the trace-normalized family is itself compatible with restriction, whether the assignment $\kappa \mapsto A_{\psi, \beta_0}(V(\kappa))$ is a functor, and — the question this campaign’s earlier working hypothesis answered wrongly — whether a single compatible choice of character can be made once and for all, for every finite field simultaneously. It cannot; the precise obstruction is Frobenius, and the precise set of compatible choices that *does* exist is named exactly.

PROVED

Theorem 8.1 (The trace-normalized family, restricted along an extension). *Let $\kappa = \mathbb{F}_{p^m} \subseteq \kappa' = \mathbb{F}_{p^{mn}}$. Then $\psi'_\zeta|_\kappa = \psi_{\zeta^n}$ (both sides characters of κ , computed with the same ζ): the restriction of κ' ’s trace-normalized character to κ is trivial exactly when $p \mid n$ (smallest instance: $\mathbb{F}_2 \subset \mathbb{F}_4$).*

Scope. This theorem is about the trace-normalized family specifically — a particular, and particularly natural, choice inside $X(\kappa)$ for every κ — and asserts nothing about arbitrary families of characters; the statement that quantifies over *all* compatible systems is Theorem 8.4 below.

Provenance. ids: D3, WH-FUNCT-a; proved in: `theory/wh-kappa-choice.md` §10 step $\langle 1 \rangle 1$; tested by: `theory/checks/funct.sections.py` (F1); reviewed in: `theory/verdicts/wh-kappa-adjudication.md`.

Proof. For $x \in \kappa$, $x^{p^m} = x$ (elements of $\mathbb{F}_{p^m}$ are exactly the fixed points of the m -th power of Frobenius), so the mn summands of $\text{Tr}_{\kappa'/\mathbb{F}_p}(x) = \sum_{k=0}^{mn-1} x^{p^k}$ repeat with period m : grouping into n blocks of m consecutive terms, each block sums to $\text{Tr}_{\kappa/\mathbb{F}_p}(x)$ (using $x^{p^{k+m}} = (x^{p^m})^{p^k} = x^{p^k}$), so $\text{Tr}_{\kappa'/\mathbb{F}_p}|_\kappa = n \cdot \text{Tr}_{\kappa/\mathbb{F}_p}$. Hence $\psi'_\zeta(x) = \zeta^{\text{Tr}_{\kappa'/\mathbb{F}_p}(x)} = \zeta^{n \cdot \text{Tr}_{\kappa/\mathbb{F}_p}(x)} = (\zeta^n)^{\text{Tr}_{\kappa/\mathbb{F}_p}(x)} = \psi_{\zeta^n}(x)$, and $\zeta^n = 1$ (so $\psi_{\zeta^n} \equiv 1$) exactly when $p \mid n$, since ζ has exact order p . $\square$

PROVED

Theorem 8.2 (The reference algebra is functorial in character-compatible embeddings). *Let $\mathcal{FF}^\pm$ be the category with objects the pairs (κ, ψ) , $\psi \in X(\kappa)$, and morphisms $(\kappa, \psi) \rightarrow (\kappa', \psi')$ the field embeddings $\iota : \kappa \hookrightarrow \kappa'$ with $\psi' \circ \iota = \psi$ (character-compatible embeddings). Then*

$$(\kappa, \psi) \longmapsto A_{\psi, \beta_0}(V(\kappa))$$

is a functor from $\mathcal{FF}^\pm$ to the category of unital $\mathbb{C}$ -algebras and injective algebra homomorphisms, respecting composition and identities.

Scope. This functor uses the reference cocycle β_0 throughout, on every object; it is not claimed that the resulting assignment is independent of this choice, nor that an analogous functor exists carrying a varying β_κ compatibly along embeddings — the reference cocycle is used here because it is a fixed, explicit stipulation (Definition 4.1), not because it is canonical (Theorem 4.4 of §4 shows at $p = 2$ it is not). Nor is this a statement about the quantum *system* — the Hilbert space or the Heisenberg group — functorially in κ , only about the algebra.

Provenance. ids: D2, D4, WH-ALG, WH-FUNCT-b; proved in: theory/wh-kappa-choice.md §10 step ⟨1⟩2; tested by: theory/checks/funct_sections.py (F5); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. For a character-compatible embedding $\iota : (\kappa, \psi) \rightarrow (\kappa', \psi')$, put $V(\iota)(a, b) := (\iota a, \iota b) : V(\kappa) \rightarrow V(\kappa')$, additive since ι is a field homomorphism. Since ι is multiplicative, $\beta_0(V(\iota)v, V(\iota)v') = \iota(a) \iota(b') = \iota(ab') = \iota(\beta_0(v, v'))$. Define $W_\beta(v) \mapsto W'_{\beta_0}(V(\iota)v)$ on basis elements of $A_{\psi, \beta_0}(V(\kappa))$. This sends distinct basis elements to distinct basis elements ($V(\iota)$ is injective, ι being a field embedding), and is multiplicative: the image of $W_{\beta_0}(v)W_{\beta_0}(v') = \psi(\beta_0(v, v'))W_{\beta_0}(v + v')$ is $\psi(\beta_0(v, v'))W'_{\beta_0}(V(\iota)(v + v'))$, while the product of the images is $\psi'(\beta_0(V(\iota)v, V(\iota)v'))W'_{\beta_0}(V(\iota)(v + v')) = \psi'(\iota(\beta_0(v, v'))W'_{\beta_0}(V(\iota)(v + v')) = \psi(\beta_0(v, v'))W'_{\beta_0}(V(\iota)(v + v'))$, using $\psi' \circ \iota = \psi$ (the compatibility condition). So the map is a unital, nonzero algebra homomorphism, hence injective by simplicity of the source (Theorem 2.8). It respects composition and identities because $\iota \mapsto V(\iota)$ does: $V(\iota' \circ \iota) = V(\iota') \circ V(\iota)$ and $V(\text{id}) = \text{id}$, directly from the definition of $V(\iota)$. $\square$

8.1 No compatible choice of character over every finite field

REFUTED

Theorem 8.3 (Refuted: a Frobenius-compatible character exists for every finite field). (Claimed, and refuted.) *A section of the forgetful functor $\mathcal{FF}^\pm \rightarrow \{\text{finite fields}\}$ exists: a choice $\psi_\kappa \in X(\kappa)$ for every finite field κ such that $\iota : \kappa \rightarrow \kappa'$ character-compatible (i.e. $\psi_{\kappa'} \circ \iota = \psi_\kappa$) for every field embedding ι — in particular, for the Frobenius self-embedding of every κ .*

Scope. This row is kept, refuted, rather than deleted, per this campaign's status discipline: a refutation carrying the surviving weaker statement is a result, not a gap to be erased. The surviving statement — exactly which obstruction blocks a section, and exactly which weaker compatible systems *do* exist — is Theorem 8.4 immediately below and Theorem 8.5 thereafter. Nothing may depend on this row.

Provenance. ids: WH-FUNCT-b-SEC; proved in: —; tested by: theory/checks/funct_sections.py (F2,F3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Refutation. Frobenius $x \mapsto x^p$ is a field embedding $\kappa \rightarrow \kappa$, so a section would in particular have to satisfy $\psi_\kappa \circ \text{Frob} = \psi_\kappa$ for every κ . Theorem 8.4 below shows this forces $\psi_\kappa = \psi_c$ for some c in the prime field $\mathbb{F}_p$, and that for $\kappa = \mathbb{F}_{p^p}$ every character of this Frobenius-invariant form restricts *trivially* to $\mathbb{F}_p$ — contradicting $\psi_{\mathbb{F}_p} \in X(\mathbb{F}_p)$, which must be nontrivial by definition. So no section exists; the smallest instance of the obstruction is $\mathbb{F}_2 \subset \mathbb{F}_4$. A previously considered route to a

section, by induction along the tower $\mathbb{F}_{p^{n!}}$, is insufficient in any case: a character of $\mathbb{F}_{p^{n!}}$ compatible at that one stage can still restrict trivially to an intermediate field, so no such induction can repair the obstruction just exhibited.

PROVED

Theorem 8.4 (The Frobenius obstruction, stated exactly). *There is no choice of $\psi_\kappa \in X(\kappa)$, one for each finite field κ , compatible with all field embeddings. Precisely: writing $\psi_c := \psi_\zeta(c \cdot)$ for $c \in \kappa$ (Theorem 6.2(a)), a character ψ_c is invariant under the Frobenius self-embedding of $\kappa = \mathbb{F}_{p^m}$ if and only if $c \in \mathbb{F}_p$; and for $\kappa = \mathbb{F}_{p^p}$, every $\mathbb{F}_p$ -valued c gives a character ψ_c that restricts trivially to the prime field $\mathbb{F}_p \subset \mathbb{F}_{p^p}$. The smallest instance is $\mathbb{F}_2 \subset \mathbb{F}_4$.*

Scope. This theorem exhibits the precise obstruction at the specific extension degree $m = p$; it does not enumerate every extension degree at which some weaker compatibility fails, and it does not address (this is Theorem 8.5) which *restricted* systems of compatible characters — over a sub-poset of embeddings, or with the “for every embedding” requirement relaxed — do exist.

Provenance. ids: D3, WH-FUNCT-a, WH-FUNCT-c; proved in: theory/wh-kappa-choice.md §10 step ⟨1⟩3; tested by: theory/checks/funct_sections.py (F2,F3); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. *Frobenius-invariance forces $c \in \mathbb{F}_p$.* Write $d := c^{1/p}$ (the unique p -th root, Frobenius being bijective on the finite field κ). Since $\text{Tr}_{\kappa/\mathbb{F}_p}(y^p) = \text{Tr}_{\kappa/\mathbb{F}_p}(y)$ for every y (Frobenius cyclically permutes the m summands of the trace, using $y^{p^m} = y$), we get $\psi_c(x^p) = \zeta^{\text{Tr}(cx^p)} = \zeta^{\text{Tr}((dx)^p)} = \zeta^{\text{Tr}(dx)} = \psi_d(x)$, i.e. $\psi_c \circ \text{Frob} = \psi_d$. Since $c \mapsto \psi_c$ is injective on κ (if $\psi_c = \psi_{c'}$ then $\text{Tr}((c - c')x) \equiv 0$ for all x , forcing $c = c'$ by the same nondegeneracy of the trace pairing used in Theorem 6.2(a)’s proof), ψ_c is Frobenius-invariant iff $d = c$; since $d^p = c$ by definition of d , substituting $d = c$ gives $c^p = c$, and conversely $c^p = c$ gives $d = c^{1/p} = c$ (Frobenius being bijective). So the condition is exactly $c^p = c$, i.e. c is a root of $X^p - X$, i.e. (this polynomial having at most p roots in a field, all p elements of $\mathbb{F}_p$ among them by Fermat’s little theorem for finite fields) $c \in \mathbb{F}_p$.

Trivial restriction at $\kappa = \mathbb{F}_{p^p}$. Here $m = p$. For $c \in \mathbb{F}_p$ and $x \in \mathbb{F}_p \subset \kappa$, every summand of $\text{Tr}_{\kappa/\mathbb{F}_p}(cx) = \sum_{k=0}^{p-1} (cx)^{p^k}$ equals cx (as $cx \in \mathbb{F}_p$ is already Frobenius-fixed), so $\text{Tr}_{\kappa/\mathbb{F}_p}(cx) = p \cdot (cx) = 0$ in characteristic p . So $\psi_c(x) = \zeta^0 = 1$ for every $x \in \mathbb{F}_p$: every Frobenius-invariant character of $\mathbb{F}_{p^p}$ — the only kind a section could assign there — restricts trivially to the prime field, contradicting the requirement $\psi_{\mathbb{F}_p} \in X(\mathbb{F}_p)$ (nontrivial by Definition 2.3) forced by compatibility along the inclusion $\mathbb{F}_p \subset \mathbb{F}_{p^p}$. At $p = 2$, $\mathbb{F}_{p^p} = \mathbb{F}_4$: the smallest instance. $\square$

8.2 The compatible systems that do exist

PROVED

Theorem 8.5 (Compatible character systems over an algebraic closure). *Fix an algebraic closure $\overline{\mathbb{F}}_p$, and consider the inclusion poset of its finite subfields. The compatible systems $(\psi_\kappa)_\kappa$ — one $\psi_\kappa \in X(\kappa)$ for every finite subfield $\kappa \subset \overline{\mathbb{F}}_p$, with $\psi_{\kappa'}|_\kappa = \psi_\kappa$ for every inclusion $\kappa \subseteq \kappa'$ of finite subfields — are **exactly** the restrictions $\psi_\kappa := \Psi|_\kappa$ of the additive characters Ψ of $(\overline{\mathbb{F}}_p, +)$ with $\Psi|_{\mathbb{F}_p} \neq 1$.*

Scope. This theorem restricts attention to the inclusion poset of finite subfields of a *fixed* algebraic closure, respecting compatibility only along inclusions of subfields, not along all abstract field embeddings between isomorphic-but-not-identical finite fields; Theorem 8.3 shows the latter, stronger requirement admits no solution at all. It says nothing about compatible choices of a polarizing cocycle β along the same poset; whether the Arf type of §4 can be chosen compatibly there is not resolved by this labbook.

Provenance. ids: D3, WH-FUNCT-d; proved in: theory/wh-kappa-choice.md §10 step ⟨1⟩4; tested by: theory/checks/funct_sections.py (F4); reviewed in: theory/verdicts/wh-kappa-adjudication.md.

Proof. Restrictions are compatible systems. Given such a Ψ , put $\psi_\kappa := \Psi|_\kappa$; compatibility along $\kappa \subseteq \kappa'$ holds by construction ($(\Psi|_{\kappa'})|_\kappa = \Psi|_\kappa$), and ψ_κ is nontrivial for *every* finite subfield κ : every such κ contains $\mathbb{F}_p$, and $\Psi|_\kappa \equiv 1$ would force $\Psi|_{\mathbb{F}_p} \equiv 1$ too, contradicting the hypothesis on Ψ .

Compatible systems arise this way. Given a compatible system (ψ_κ) , define $\Psi(x) := \psi_\kappa(x)$ for any finite subfield $\kappa \ni x$ (every $x \in \overline{\mathbb{F}_p}$ lies in some finite subfield, $\overline{\mathbb{F}_p}$ being their union). This is well defined: if x lies in two finite subfields κ, κ' , both lie inside a common finite subfield κ'' (e.g. their compositum, again finite), and compatibility along $\kappa \subseteq \kappa''$ and $\kappa' \subseteq \kappa''$ gives $\psi_\kappa(x) = \psi_{\kappa''}(x) = \psi_{\kappa'}(x)$. Ψ is additive: any two elements lie in a common finite subfield, where Ψ agrees with the additive character ψ_κ of that subfield. And $\Psi|_{\mathbb{F}_p} = \psi_{\mathbb{F}_p} \neq 1$, by Definition 2.3's requirement on every $\psi_\kappa \in X(\kappa)$ applied to $\kappa = \mathbb{F}_p$.

The two constructions are mutually inverse, and the condition “nontrivial on every finite subfield” (needed termwise, since each $\psi_\kappa \in X(\kappa)$ by hypothesis) is equivalent to “nontrivial on the prime field alone”: the forward implication is immediate ($\mathbb{F}_p$ is itself one of the subfields), and the reverse is exactly what the first paragraph showed. $\square$

Remark (What this section does not claim). Theorem 8.2 is functoriality of the *algebra* in character-compatible embeddings, for the fixed reference cocycle β_0 — not functoriality of the quantum system (Heisenberg group, Hilbert space) itself, and not a claim that any compatible choice of polarizing cocycle along embeddings exists. Whether the Arf type of Theorem 4.7 can be chosen compatibly along field embeddings at $p = 2$ is recorded as an open question by this campaign's source shard, not resolved here.

9 Weyl–Heisenberg quantization over finite local rings

This section records the first extension from finite fields to finite commutative local rings. In the preferred three-stage pipeline (P1), the ring first produces the symplectic module R^2 ; a character and a polarizing cocycle then produce its Weyl algebra and Heisenberg group; finally the irreducible unitary representations are classified. The decisive condition is that the chosen character generate the character module. At odd residue characteristic this gives a single irreducible representation of dimension $|R|$.

There is a genuine fork at the first stage. Route A, used here, takes $R \oplus R$ with an R -valued alternating form and then composes with a chosen additive character. Route B would take $R \oplus \widehat{R}$ with its canonical circle-valued Pontryagin pairing; it needs no character to become nondegenerate. Route B and the comparison of the physics retained by the two routes are deferred; no result below ranks the routes.

9.1 The local datum and its phase space

Definition 9.1 (Finite commutative local ring datum). A finite local ring datum is $(R, \mathfrak{m}, \kappa, q)$ with R a finite commutative unital local ring with $1 \neq 0$, $\mathfrak{m}$ its unique maximal ideal, $\kappa := R/\mathfrak{m}$, and $q := |\kappa|$; write $R^\times$ for its unit group. For an ideal $I \subseteq R$, put

$$\text{Ann}(I) := \{r \in R : rI = 0\}, \quad \text{soc}(R) := \text{Ann}(\mathfrak{m}).$$

These are stipulations. Nilpotence of $\mathfrak{m}$, nonvanishing of the socle, and the other finite-local structure properties are established below, not assumed here. The definition is uniform in the residue characteristic.

Scope. Only finite commutative unital local rings with $1 \neq 0$ are defined here. No Frobenius, characteristic, principal-ideal, or chain-ring hypothesis is part of the datum.

Provenance. ids: D12; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

Definition 9.2 (Characters and generating characters). Let

$$\widehat{R} := \text{Hom}((R, +), \mathbb{C}^\times), \quad X(R) := \widehat{R} \setminus \{1\}.$$

For $\psi \in X(R)$ and $u \in R$, put $\psi_u(x) := \psi(ux)$. Define

$$I_\psi := \sum \{I \subseteq R : I \text{ is an ideal and } I \subseteq \ker \psi\}, \quad \text{Gen}(R) := \{\psi \in X(R) : I_\psi = 0\}.$$

Thus I_ψ is the largest ideal contained in $\ker \psi$, and a member of $\text{Gen}(R)$ is a *generating character*. This definition is uniform in the residue characteristic; existence is not stipulated.

Scope. The definition includes every nontrivial additive character, not only a trace-normalized family. It makes no assertion that generating characters exist for an arbitrary finite local ring.

Provenance. ids: D13; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

Definition 9.3 (Symplectic module and phase perpendicularity). Put $V(R) := R \oplus R$ and

$$\omega((a, b), (a', b')) := ab' - a'b.$$

Given $\psi \in X(R)$, write

$$B_\psi(v, w) := \psi(\omega(v, w)), \quad L^{\perp_\psi} := \{v \in V(R) : B_\psi(v, l) = 1 \text{ for every } l \in L\}$$

for an R -submodule $L \subseteq V(R)$. A *Lagrangian* is an R -submodule satisfying $L = L^{\perp_\psi}$, and $\text{rad}(B_\psi) := V(R)^{\perp_\psi}$. Bilinearity, strong alternation, R -nondegeneracy of ω , and nondegeneracy of B_ψ are claims, not stipulations; the character-valued pairing B_ψ , not R -nondegeneracy alone, is load-bearing. This definition is uniform in the residue characteristic.

Scope. Perpendicularity is defined only for R -submodules of R^2 and is measured through the chosen character. No freeness condition is imposed on a Lagrangian.

Provenance. ids: D14; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

Definition 9.4 (Admissible polarizing cocycles over a local ring). For the symplectic module above, set

$$\begin{aligned} \text{Adm}(\omega) &:= \{\beta : V(R) \times V(R) \rightarrow R : \beta \text{ is } R\text{-bilinear and } \beta - \beta^T = \omega\}, \\ \text{Sym}_R(V(R)) &:= \{s : V(R) \times V(R) \rightarrow R : s \text{ is } R\text{-bilinear and } s = s^T\}. \end{aligned}$$

A *polarizing cocycle* is a named choice $\beta \in \text{Adm}(\omega)$. The reference member is

$$\beta_0((a, b), (a', b')) := ab'.$$

This is the nonsymmetrized convention: no inverse of 2 is assumed. The definition is uniform in the residue characteristic.

Scope. The reference member is a stipulated coordinate formula, not a claim that the torsor has a canonical point. The symmetrized member $\omega/2$ is used only under the explicit hypothesis $2 \in R^\times$.

Provenance. ids: D15; proved in: —; tested by: —; reviewed in: theory/verdicts/fcr-local-r1.md.

Definition 9.5 (Local Weyl algebra, Heisenberg group, and models). For the data above define

$$A_{\psi,\beta}(V(R)) := \bigoplus_{v \in V(R)} \mathbb{C} W_{\beta}(v), \quad W_{\beta}(v)W_{\beta}(v') := \psi(\beta(v, v'))W_{\beta}(v + v'),$$

and

$$H_{\beta}(R) := R \times V(R), \quad (t, v)(t', v') := (t + t' + \beta(v, v'), v + v').$$

For an R -submodule L , put $A_L := \text{span}_{\mathbb{C}}\{W_{\beta}(l) : l \in L\}$. Given a unital algebra character $\chi : A_L \rightarrow \mathbb{C}$, put

$$M_{L,\chi} := A_{\psi,\beta}(V(R)) \otimes_{A_L} \mathbb{C}_{\chi}.$$

The fixed reference model is $\ell^2(R) = \bigoplus_{y \in R} \mathbb{C} e_y$, with $\{e_y\}$ orthonormal,

$$X(a)e_y := e_{y+a}, \quad Z(b)e_y := \psi(by),$$

and, exactly,

$$W_{\beta_0}(a, b) := Z(-b)X(a), \quad W_{\beta_0}(a, b)e_y = \psi(-b(y + a))e_{y+a}.$$

The sign and ordering are stipulations. Every displayed formula is uniform in the residue characteristic and uses no half.

Scope. The algebra and group depend on the named cocycle, while the algebra and representation also depend on the named character. A submodule and its character label a realization; neither is added to the underlying algebra datum.

Provenance. ids: D16; proved in: —; tested by: theory/checks/fcr_local_check.py (G4); reviewed in: theory/verdicts/fcr-local-r1.md.

9.2 Generating characters and the collapse boundary

PROVED

Theorem 9.6 (Local Frobenius criterion and the unit torsor). *Let R be a finite commutative local ring with maximal ideal $\mathfrak{m}$ and residue field κ . Then*

$$\text{Gen}(R) \neq \emptyset \iff \dim_{\kappa} \text{soc}(R) = 1.$$

If $\psi \in \text{Gen}(R)$, then ψ_u is generating exactly when $u \in R^{\times}$, and $R^{\times}$ acts freely and transitively on $\text{Gen}(R)$.

Scope. No odd-characteristic hypothesis is used. For rings whose socle has κ -dimension greater than one, the theorem asserts nonexistence of a generating character; it does not propose a replacement quantization.

Provenance. ids: D12, D13, FCR-GEN; proved in: theory/fcr-local.md §1; tested by: theory/checks/fcr_local_check.py (G1,G8); reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. Finiteness makes R Artinian, so $\mathfrak{m}$ is nilpotent. Wood's module socle $S({}_R R)$ equals $\text{Ann}(\mathfrak{m})$: every simple ideal J has $\mathfrak{m}J = 0$, since $\mathfrak{m}J = J$ would contradict nilpotence; conversely, if $0 \neq x \in \text{Ann}(\mathfrak{m})$, then $Rx = \kappa x$ is one-dimensional over κ and is simple. Wood's local Frobenius condition is therefore $\kappa \cong \text{soc}(R)$.

Wood's character-module theorem and kernel-ideal criterion identify this condition with the existence of ψ whose kernel contains no nonzero ideal, namely $I_{\psi} = 0$. For such a ψ , multiplication by a unit preserves $I_{\psi} = 0$, whereas a nonunit lies in $\mathfrak{m}$ and kills the nonzero socle. The character-module isomorphism then shows that every generating character is uniquely ψ_u for a unit u . $\square$

PROVED

Theorem 9.7 (Radical of the character-valued symplectic pairing). *Let R be a finite commutative local ring and let $\psi \in X(R)$. The form ω on R^2 is R -bilinear, strongly alternating, and R -nondegenerate, while*

$$\text{rad}(B_\psi) = I_\psi \oplus I_\psi.$$

Consequently B_ψ is nondegenerate if and only if $\psi \in \text{Gen}(R)$.

For $R = \mathbb{F}_3[x, y]/(x, y)^2$, write $\psi_{a,b,c}(r + sx + ty) = \zeta_3^{ar+bs+ct}$. The two nontrivial characters with $(b, c) = (0, 0)$ have $I_\psi = (x, y)$, and each of the remaining 24 has $I_\psi = \mathbb{F}_3(cx - by)$. Thus all 26 nontrivial characters collapse the pairing through a proper quotient.

Scope. The radical formula is uniform in the residue characteristic. The square-zero-ring census records the obstruction at one designated seed only; it does not construct a quantum system for a degenerate pairing and does not state a general quotient-collapse theorem.

Provenance. ids: D12, D13, D14, FCR-RAD; proved in: theory/fcr-local.md §2; tested by: theory/checks/fcr_local_check.py (G2,G3); exhaustive 26-character seed census in G1; reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. Testing $\omega((a, b), -)$ against $(0, r)$ and $(r, 0)$ shows that a radical vector must have $aR, bR \subseteq \ker \psi$, hence $a, b \in I_\psi$. The reverse inclusion follows because I_ψ is an ideal. Bilinearity and strong alternation are immediate from the displayed determinant formula, and tests against $(0, 1)$ and $(1, 0)$ prove R -nondegeneracy. In the square-zero example every subspace of (x, y) is an ideal; restricting the character to that two-dimensional space gives exactly the displayed kernel line, or all of (x, y) when the restriction vanishes. $\square$

9.3 Odd residue characteristic: algebra and representation

PROVED

Theorem 9.8 (Weyl commutation over a finite local ring). *Let R be a finite commutative local ring, let $\psi \in X(R)$, and let $\beta \in \text{Adm}(\omega)$. For every $v, v' \in V(R)$,*

$$W_\beta(v)W_\beta(v') = \psi(\omega(v, v'))W_\beta(v')W_\beta(v),$$

and

$$W_\beta(u)W_\beta(v)W_\beta(u)^{-1} = \psi(\omega(u, v))W_\beta(v).$$

Scope. The identities are uniform in the residue characteristic and do not require ψ to be generating. They do not by themselves imply simplicity of the Weyl algebra.

Provenance. ids: D14, D15, D16, FCR-COMM; proved in: theory/fcr-local.md §3; tested by: theory/checks/fcr_local_check.py (G2,G4); reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. The two products have the same basis term $W_\beta(v + v')$. Their scalar ratio is $\psi(\beta(v, v') - \beta(v', v)) = \psi(\omega(v, v'))$; conjugation gives the second identity. $\square$

PROVED[†]

Theorem 9.9 (Odd-characteristic transport of polarizing cocycles). *Let R be a finite commutative local ring with $2 \in R^\times$. Then $\text{Adm}(\omega)$ is a torsor under $\text{Sym}_R(V(R))$ and has $|R|^3$ elements. Its unique antisymmetric member is $\omega/2$. For every $\beta \in \text{Adm}(\omega)$ and every symmetric s , the map*

$$\varphi_s(t, v) := \left(t + \frac{s(v, v)}{2}, v \right)$$

is an isomorphism $H_\beta(R) \rightarrow H_{\beta+s}(R)$ fixing the centre pointwise.

Scope. The condition $2 \in R^\times$ is essential to the displayed representative and transport. No symmetry group of the ring-side phase space is analyzed here.

Provenance. ids: D14, D15, D16, FCR-BETA-ODD; proved in: theory/fcr-local.md §4; tested by: theory/checks/fcr_local_check.py (G9); reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. The difference of two admissible forms is symmetric, and adding a symmetric form preserves admissibility. A symmetric form on R^2 is freely determined by three entries. Antisymmetry and admissibility give $2\beta = \omega$, hence the unique member $\omega/2$. Finally, for $f(v) = s(v, v)/2$, symmetry gives $f(v + w) = f(v) + f(w) + s(v, w)$, which is exactly the coboundary identity needed for φ_s to respect the two group laws. $\square$

PROVED [†]

Theorem 9.10 (Central simplicity and finite Stone–von Neumann theorem). *Let R be a finite commutative local ring, let $\psi \in \text{Gen}(R)$, let $\beta \in \text{Adm}(\omega)$, and suppose $2 \in R^\times$. Then $A_{\psi, \beta}(V(R))$ is simple, has centre $\mathbb{C}1$, has complex dimension $|R|^2$, and is noncanonically isomorphic to $M_{|R|}(\mathbb{C})$. The group $H_\beta(R)$ has exactly one irreducible unitary representation with central character ψ , up to unitary equivalence; its dimension is $|R|$, unitary intertwiners are unique up to $U(1)$, and every complex algebra automorphism of the Weyl algebra is inner.*

Scope. The condition $2 \in R^\times$ is used to transport an arbitrary cocycle to the fixed matrix model. The matrix isomorphism depends on choices of realization and basis and is not part of the algebra datum. No frame-preserving automorphism group is computed.

Provenance. ids: D12–D16, FCR-ALG, FCR-SVN; proved in: theory/fcr-local.md §5–§6; tested by: theory/checks/fcr_local_check.py (G4, G5, G6), including the G6 arena mutation; reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. Averaging conjugation by all Weyl basis elements kills every nonzero basis coefficient because the pairing is nondegenerate. It follows that the centre is scalar and that every nonzero two-sided ideal contains 1. The reference operators $Z(-b)X(a)$ give a nonzero representation on $\ell^2(R)$; simplicity makes it injective and the equal dimensions make it onto $\text{End}_{\mathbb{C}}(\ell^2(R))$. The preceding transport handles arbitrary β .

Modules with central character ψ are precisely modules over this full matrix algebra. Its column module is the unique simple module and has dimension $|R|$. Schur’s argument makes intertwiners scalar; polar rescaling gives the $U(1)$ statement. Twisting the column module by an algebra automorphism shows that the automorphism is conjugation by an invertible operator. $\square$

PROVED [†]

Theorem 9.11 (Lagrangians and Schrödinger models over a local ring). *Let R be a finite commutative local ring, let $\psi \in \text{Gen}(R)$, let $\beta \in \text{Adm}(\omega)$, and suppose $2 \in R^\times$. Every Lagrangian $L \subseteq V(R)$ has $|L| = |R|$. The coordinate axes are Lagrangian, and for every ideal $I \subseteq R$ so is*

$$I \oplus \text{Ann}(I).$$

If R is not a field, then

$$L_* := \text{soc}(R) \oplus \mathfrak{m}$$

is a nonfree Lagrangian. Every algebra character $\chi : A_L \rightarrow \mathbb{C}$ produces a simple induced module $M_{L, \chi}$ of dimension $|R|$, and all these modules are isomorphic.

Scope. This is a catalogue containing a general ideal family, not a classification of all Lagrangians over every local ring. In particular, it does not claim that every Lagrangian is free or belongs to the displayed ideal family.

Provenance. ids: D12–D16, FCR-POL; proved in: theory/fcr-local.md §7; tested by: theory/checks/fcr_local_check.py (G7), full submodule census at the two order-nine seeds; reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. Character orthogonality in the two orders gives $|L| |L^{\perp\psi}| = |R|^2$, so a self-perpendicular module has order $|R|$. Annihilator duality gives $\text{Ann}(\text{Ann}(I)) = I$, hence the ideal family. For a nonfield local ring, $\text{Ann}(\mathfrak{m}) = \text{soc}(R)$ and $\text{Ann}(\text{soc}(R)) = \mathfrak{m}$, so L_* belongs to that family. Its quotient by $\mathfrak{m}L_*$ needs at least two generators, whereas a free module of order $|R|$ would have rank one.

The restriction of β to a Lagrangian is symmetric, and $\chi_0(W(l)) = \psi(-\beta(l, l)/2)$ is an algebra character. The Weyl basis makes the algebra right-free over A_L on representatives for $V(R)/L$, so the induced module has dimension $|R|$; uniqueness of the simple module then proves that all labels give isomorphic realizations. $\square$

PROVED [†]

Proposition 9.12 (The two presentation data and bare-algebra ambiguity). *Let R be a finite commutative local ring with $2 \in R^\times$ and $\text{Gen}(R) \neq \emptyset$. Beyond R , the presentation uses exactly a generating character $\psi \in \text{Gen}(R)$ and a cocycle $\beta \in \text{Adm}(\omega)$. Distinct generating characters give inequivalent representations of the same group $H_\beta(R)$ because their central characters differ. All resulting bare algebras are isomorphic, but the torsor of bare-algebra isomorphisms has no point fixed by every target automorphism.*

Scope. The last assertion concerns all bare-algebra isomorphisms. It neither computes nor identifies a frame-preserving isomorphism torsor and makes no claim about a ring-side Weil or metaplectic symmetry.

Provenance. ids: D12–D16, FCR-CHOICE; proved in: theory/fcr-local.md §8; tested by: theory/checks/fcr_local_check.py (G8, G9); reviewed in: theory/verdicts/fcr-local-r1.md.

9.4 Clause-level return to an odd finite field

PROVED [†]

Theorem 9.13 (Clause-level compatibility with the odd-field system). *Let $R = \kappa$ be a finite field of odd characteristic and put $q := |\kappa|$. Specialization recovers on the nose:*

- (i) *the nontrivial-character set and its $\kappa^\times$ -torsor of order $q - 1$;*
- (ii) *bilinearity, strong alternation, and nondegeneracy of ω , together with the zero radical of $\psi \circ \omega$;*
- (iii) *the Weyl commutator and conjugation identities;*
- (iv) *simplicity, scalar centre, dimension q^2 , and a noncanonical $M_q(\mathbb{C})$ realization;*
- (v) *the unique irreducible unitary representation of dimension q , its $U(1)$ -unique intertwiners, and innerness of algebra automorphisms;*
- (vi) *exactly $q + 1$ Lagrangian lines, q algebra characters per line, $q(q + 1)$ model labels, and one module-isomorphism class;*
- (vii) *the torsor of admissible cocycles of order q^3 , its unique antisymmetric member $\omega/2$, and the centre-fixing transports;*
- (viii) *the two named presentation data, the distinct-central-character obstruction, and isomorphism of the resulting bare algebras.*

The specialized definitions give the same $V(\kappa)$, form, admissibility equation, Weyl and group laws, and fixed operator $W_{\beta_0} = Z(-b)X(a)$ with the same sign.

Scope. This is deliberately a clause-level comparison. It does not compare the field theory's frame-preserving isomorphism torsor of order $q^2|SL_2(\kappa)|$; the ring-side frame-preserving analysis is future work. It also does not compare the field theorem's cocycle or automorphism-group clauses, the $SL_2(\kappa)$ -invariance of $\omega/2$, or the particular polarization-dependent matrix construction used in the field proof.

Provenance. ids: D12–D16, FCR-GEN, FCR-RAD, FCR-COMM, FCR-BETA-ODD, FCR-ALG, FCR-SVN, FCR-POL, FCR-CHOICE, FCR-REG; proved in: theory/fcr-local.md §9; tested by: theory/checks/fcr_local_check.py (G10), exact counts, form and radical tables, fixed matrices, rank, commutant, lines, half-census, and group profile; reviewed in: theory/verdicts/fcr-local-r1.md.

Proof. For a field, $\mathfrak{m} = 0$, $\text{soc}(R) = R$, and the only ideals are 0 and R , so every nontrivial character is generating. Substitution in the preceding results proves every listed clause. A self-perpendicular subspace of the nondegenerate two-dimensional space is a line; nonzero vectors modulo $\kappa^\times$ give $(q^2 - 1)/(q - 1) = q + 1$ lines, and each line has q additive characters. No excluded symmetry or frame clause follows from these calculations. $\square$

10 The five smallest local rings, worked in full

The three smallest orders of finite commutative local rings are 2, 3, and 4, and order 4 is a three-way tie. This section exercises every definition of Section 9 on the resulting five rings

$$\mathbb{F}_2, \quad \mathbb{F}_3, \quad \mathbb{F}_4, \quad \mathbb{Z}/4, \quad \mathbb{F}_2[\varepsilon]/(\varepsilon^2),$$

computing their Weyl–Heisenberg groups and complete irreducible representation catalogues. Every number in this section is verified by the exact-arithmetic census checker `theory/checks/small_rings_cat` written and run by an independent verifier lane that recomputed the orchestrator's catalogue blind; the checker runs green and carries eight mutation modes, each of which fails at its registered gate. Statements here are census facts about these five rings; no generality beyond them is claimed except where a claim identifier is cited.

10.1 The data, ring by ring

All five rings are Frobenius: the socle is one-dimensional over the residue field, so generating characters exist and form a single free orbit of the unit group (the existence criterion and torsor structure of Section 9).

R	$\mathfrak{m}$	κ	$\text{soc}(R)$	$ R^\times = \text{Gen}(R) $	reference ψ
$\mathbb{F}_2$	0	$\mathbb{F}_2$	R	1	$(-1)^x$
$\mathbb{F}_3$	0	$\mathbb{F}_3$	R	2	ζ_3^x
$\mathbb{F}_4$	0	$\mathbb{F}_4$	R	3	$(-1)^{\text{Tr}(x)}$
$\mathbb{Z}/4$	(2)	$\mathbb{F}_2$	(2)	2	i^x
$\mathbb{F}_2[\varepsilon]$	(ε)	$\mathbb{F}_2$	(ε)	2	$(-1)^{a+b}$ for $x = a + b\varepsilon$

Provenance. ids: FCR-GEN; proved in: theory/fcr-local.md §1; tested by: `small_rings_catalogue_check.py` (ring-data gates); reviewed in: theory/lanes/smallrings/verify/VERIFICATION.md.

10.2 The Heisenberg groups are unitriangular matrix groups

Proposition 10.1 (Matrix form of the reference-cocycle Heisenberg group). *For every finite commutative unital ring R , the map*

$$(t, (a, b)) \longmapsto \begin{pmatrix} 1 & a & t \\ 0 & 1 & b \\ 0 & 0 & 1 \end{pmatrix}$$

is an isomorphism from the Heisenberg group of the reference cocycle $\beta_0((a, b), (a', b')) = ab'$ onto the unitriangular group $\text{UT}_3(R)$.

Proof. Multiplying the two displayed matrices gives upper-right entry $t+t'+ab'$, which is exactly the group law of the Heisenberg group of β_0 ; the map is a bijection by construction. $\square$

Scope. The identification is specific to the reference cocycle: at residue characteristic 2 other admissible cocycles can give non-isomorphic groups (the quaternion type over $\mathbb{F}_2$ below), and those are not unitriangular in this way.

Provenance. ids: D16; proved in: this section; tested by: small_rings_catalogue_check.py (cocycle gate); reviewed in: theory/lanes/smallrings/verify/VERIFICATION.md.

The five groups, exactly:

R	$ H $	centre	exponent	order profile	identification
$\mathbb{F}_2$	8	$\mathbb{Z}/2$	4	1:1, 2:5, 4:2	dihedral D_4
$\mathbb{F}_3$	27	$\mathbb{Z}/3$	3	1:1, 3:26	extraspecial 3_+^{1+2}
$\mathbb{F}_4$	64	$(\mathbb{Z}/2)^2$	4	1:1, 2:27, 4:36	characteristic-two Heisenberg
$\mathbb{Z}/4$	64	$\mathbb{Z}/4$	8	1:1, 2:7, 4:40, 8:16	$\text{UT}_3(\mathbb{Z}/4)$
$\mathbb{F}_2[\varepsilon]$	64	$(\mathbb{Z}/2)^2$	4	1:1, 2:31, 4:32	$\text{UT}_3(\mathbb{F}_2[\varepsilon])$

The three order-64 groups are pairwise non-isomorphic: $\mathbb{Z}/4$ is separated by its cyclic centre and its sixteen elements of order 8, while $\mathbb{F}_4$ and $\mathbb{F}_2[\varepsilon]$ share centre and exponent but differ in involution count. The order-8 elements over $\mathbb{Z}/4$ come from the square law $(t, v)^2 = (2t + \beta_0(v, v), 2v)$: the fourth power is $(2\beta_0(v, v), 0)$, nonzero exactly when ab is odd. Over $\mathbb{F}_2[\varepsilon]$ the identity $2v = 0$ kills this mechanism: the jet direction deepens the ring without deepening the clock. Mixed versus equal characteristic is thus visible in the bare group.

Remark. At $\mathbb{F}_3$ the group is independent of the cocycle choice (the odd-characteristic transport of Section 9). At $\mathbb{F}_2$ the eight admissible cocycles split six-to-two into a dihedral type and a quaternion type, separated by the Arf invariant of $Q_\beta(v) = \beta(v, v)$; the reference cocycle is of dihedral type. For $\mathbb{Z}/4$ and $\mathbb{F}_2[\varepsilon]$ the classification of the 64-element cocycle torsor is open and is the target of the running residue-characteristic-two increment.

Provenance. ids: WH-BETA-d, FCR-BETA-ODD; proved in: theory/wh-kappa-choice.md §6; theory/fcr-local.md §4; tested by: beta_census.py; small_rings_catalogue_check.py; reviewed in: theory/verdicts/wh-kappa-adjudication.md.

10.3 The complete representation catalogues

The commutator subgroup is the full centre, so the abelianization is the phase module $R \oplus R$, and irreducible representations are stratified by the character of the centre, $\psi_u = \psi(u \cdot)$ for $u \in R$.

Proposition 10.2 (Catalogue census for the five smallest local rings). *For each of the five rings above and each $u \in R$, the number of irreducible unitary representations of the Heisenberg group with central character ψ_u is $|\text{Ann}(u)|^2$, and each has dimension $|R|/|\text{Ann}(u)|$. In particular the total count of irreducibles equals the number of conjugacy classes,*

$$\#\{\text{classes}\} = |R|^2 + \sum_{u \neq 0} |\text{Ann}(u)|^2 \quad (5, 11, 19, 22, 22 \text{ respectively}),$$

and the sum of squared dimensions equals $|R|^3$.

Scope. A census statement about these five rings only, established by exhaustive computation; the same stratification pattern for a general finite commutative local ring is expected but not claimed here — its general proof belongs to the planned non-Frobenius/collapse increment.

Provenance. ids: (census); proved in: small_rings_catalogue_check.py (class-count and catalogue gates); tested by: same checker, mutation modes red-class-count, red-catalogue; reviewed in: theory/lanes/smallrings/verify/VERIFICATION.md.

The three strata carry distinct physics:

- *Bottom*, $u = 0$: the $|R|^2$ characters of the abelianization — the classical shadow.
- *Top*, $u \in R^\times$: one Schrödinger representation per generating character, of dimension $|R|$, in the fixed model $W(a, b) = Z(-b)X(a)$ on $\ell^2(R)$: the qubit Pauli pair for $\mathbb{F}_2$; the qutrit clock-shift for $\mathbb{F}_3$; two $\mathbb{F}_4$ -labelled qubits (three inequivalent copies, one per generating character); the ququart clock-shift $Z = \text{diag}(1, i, -1, -i)$, $ZX = iXZ$ for $\mathbb{Z}/4$ (two copies); and the jet-paired four-dimensional representation for $\mathbb{F}_2[\varepsilon]$ (two copies). Irreducibility is pinned by exact trace-orthogonality of the $|R|^2$ Weyl operators.
- *Middle*, $u \in \mathfrak{m} \setminus 0$ (only the two thickened rings): four two-dimensional representations each, with central character supported on the socle. These are qubit representations of the collapsed quotient $R/\text{Ann}(u) \cong \mathbb{F}_2$ twisted by characters of the radical of the degenerate phase pairing. The verifier's decomposition of the four-dimensional model with this central character found exactly two inequivalent two-dimensional blocks, each of multiplicity one — the model contains two of the four classes.

A ring is thickened exactly when its catalogue has a middle stratum: the representation theory detects the nilpotents. Two honest cautions from the blind verification: $\mathbb{F}_4$ has *three* top-stratum representations, not two; and as bare dimension-multiplicity tables the $\mathbb{Z}/4$ and $\mathbb{F}_2[\varepsilon]$ catalogues coincide ($16 \times \dim 1$, $4 \times \dim 2$, $2 \times \dim 4$) — the catalogue separates the two rings only through its central-character labelling and the underlying groups, not through the numbers alone.

10.4 Lagrangian labels and models

R	Lagrangians	free	non-free	model labels
$\mathbb{F}_2$	3	3	0	6
$\mathbb{F}_3$	4	4	0	12
$\mathbb{F}_4$	5	5	0	20
$\mathbb{Z}/4$	7	6	1	28
$\mathbb{F}_2[\varepsilon]$	7	6	1	28

The fields show the $q + 1$ lines of the projective line; the thickened rings show the six free lines of $\mathbb{P}^1(R)$ plus the non-free witness $\text{soc}(R) \oplus \mathfrak{m}$ of Section 9. All labels with a fixed generating central character present one representation, but the non-free label is a genuinely

different presentation: position and momentum are each sampled at depth $\mathfrak{m}$, a mixed frame with no field analogue.

Provenance. ids: FCR-POL; proved in: theory/fcr-local.md §7; tested by: small_rings_catalogue_check.py (Lagrangian gate, mutation mode red-drop-nonfree); reviewed in: theory/verdicts/fcr-local-adjudication.md.